

2016.0 RANGE ROVER (LG), 309-00

EXHAUST SYSTEM – TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLE OF OPERATION

For a detailed description of the Exhaust System, refer to the relevant Description and Operation section of the workshop manual.

REFER to: [Selective Catalyst Reduction](#) (309-00A Exhaust System - TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- If a control module or a component is suspect and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval programme is in operation, prior to the installation of a new module/component.
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern

2. Visually inspect for obvious signs of damage and system integrity

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Leaks ▪ Metal fatigue ▪ Correct installation ▪ Selective catalyst reduction muffler ▪ Muffler(s) ▪ Joints ▪ Mountings ▪ Clearance around components ▪ Diesel exhaust fluid injector ▪ Diesel exhaust fluid line ▪ Diesel exhaust fluid tank ▪ Diesel exhaust fluid filler assembly ▪ Diesel exhaust fluid level	▪ Battery ▪ Engine junction box ▪ Rear junction box ▪ Fuses ▪ Wiring harnesses and connectors ▪ Engine control module ▪ Sensor(s) ▪ Actuator(s) ▪ DEF relay

- Diesel exhaust fluid quality

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index
5. Check DDW for open campaigns. Refer to the corresponding bulletins and SSMs which may be valid for the specific customer complaint and carry out the recommendations as required

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
▪ Engine malfunction lamp illuminated ▪ Or ▪ Reagent indicator (17MY on)	▪ Selective catalyst reduction system failure ▪ Diesel exhaust fluid level too low ▪ Power or ground failure ▪ Diesel exhaust fluid pressure too low ▪ Diesel exhaust fluid pressure too high ▪ Diesel exhaust fluid injection pump ▪ Diesel exhaust fluid injector ▪ Diesel exhaust fluid relay ▪ Diesel exhaust fluid heater control unit ▪ Pre selective catalytic reduction NOx sensor ▪ Pre selective catalyst reduction Nox sensor module ▪ Post selective catalyst reduction NOx sensor ▪ Post selective catalyst reduction Nox sensor module	▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index ▪ Check and correct the diesel exhaust fluid level ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Refer to the electrical circuit diagrams and check sensor,

	▪ Post selective catalyst reduction soot sensor ▪ Post selective catalyst reduction soot sensor module ▪ Diesel sub net CAN communication bus	relays/fuses and control module power feeds and grounds ▪ Check system for correct installation of components ▪ Check system mechanical and fluid integrity ▪ Check system for mechanical integrity ▪ Check system for leakage ▪ Check system for blockage
▪ Engine will not crank	▪ Vehicle battery failure ▪ Diesel exhaust fluid level too low ▪ Harness failure - Wiring integrity	▪ Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and correct the diesel exhaust fluid level ▪ Using the manufacturer approved diagnostic system carry out any required routines
▪ Message warning	▪ Selective catalyst reduction system failure ▪ A monitor of the historical diesel exhaust fluid consumption rate, allied with a direct reading of the diesel exhaust fluid level in the tank, predicts that the distance to emptying the tank is less than 2400km	▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index ▪ Check and correct the diesel exhaust fluid level

CONTROL MODULE CAVITY

CONTROL MODULE CAVITY	DESCRIPTION	DTC
Circuit reference I_T_RAULS Signal	PWM input to the engine control module	▪ P203B-02 ▪ P203B-16 ▪ P203B-17 ▪ P203B-31 ▪ P203C-00 ▪ P203D-00 ▪ P205B-84 ▪ P205B-85 ▪ P205C-16 ▪ P205D-17 ▪ P205E-64
Circuit reference RAPM	PWM output from the engine control module	▪ P208A-13 ▪ P208B-4B ▪ P208C-11 ▪ P208D-12
Circuit reference RAMV_L	PWM output from the engine control module	▪ P202E-4B ▪ P2047-13 ▪ P2048-11 ▪ P2049-12
Circuit reference RAMV_H	PWM output from the engine control module	▪ P202E-4B

		▪ P2047-13 ▪ P2048-11 ▪ P2049-12
Circuit reference CAN_3_H	Diesel sub net CAN communication bus (High)	▪ P2200-13 ▪ P2200-14 ▪ P2201-28 ▪ P2201-2F ▪ P2202-16 ▪ P2203-17 ▪ P2209-01 ▪ P220A-01 ▪ P220B-01 ▪ P225C-85 ▪ P225D-84 ▪ P225E-85 ▪ P225F-84 ▪ P229E-13 ▪ P229E-14 ▪ P229F-28 ▪ P229F-2F ▪ P22A0-16 ▪ P22A1-17 ▪ P22A7-01 ▪ P22FA-27 ▪ P22FB-64 ▪ P22FB-84 ▪ P22FB-85

		85 ▪ P22FE-2A ▪ P22FE-64 ▪ P22FE-84 ▪ P22FE-85 ▪ U029D-87 ▪ U029E-87 ▪ U02A5-87 ▪ U05A6-08 ▪ U3009-00 ▪ U300C-00 ▪ U300F-00
Circuit reference CAN_L_3	Diesel sub net CAN communication bus (Low)	▪ P2200-13 ▪ P2200-14 ▪ P2201-28 ▪ P2201-2F ▪ P2202-16 ▪ P2203-17 ▪ P2209-01 ▪ P220A-01 ▪ P220B-01 ▪ P225C-85 ▪ P225D-84 ▪ P225E-85 ▪ P225F-84 ▪ P229E-13 ▪ P229E-14

		▪ P229F-28 ▪ P229F-2F ▪ P22A0-16 ▪ P22A1-17 ▪ P22A7-01 ▪ P22FA-27 ▪ P22FB-64 ▪ P22FB-84 ▪ P22FB-85 ▪ P22FE-2A ▪ P22FE-64 ▪ P22FE-84 ▪ P22FE-85 ▪ U029D-87 ▪ U029E-87 ▪ U02A5-87 ▪ U05A6-08 ▪ U3009-00 ▪ U300C-00 ▪ U300F-00
Circuit reference SCRMRLY	PWM output from the engine control module	▪ P21C7-13 ▪ P21C8-11 ▪ P21C9-

		■ P2107-12
Circuit reference RAPP	PWM output from the engine control module	■ P20FA-13 ■ P20FC-11 ■ P20FD-12

DIAGNOSTIC ROUTINES

ROUTINE NUMBER	ROUTINE DESCRIPTION	ROUTINE ACTION
0x0406	Reset all adaptions	Value 0x0F - To Reset Selective Catalyst Reductant (SCR) Inducement System Start Inhibit
0x4026	Reset Specified Information	Value 0x2C – To Reset Selective Catalyst Reductant (SCR) Quality Monitor
0x4076	Selective Catalytic Reductant Preparation	This routine forms part of the DEF prime routine to be carried out at end of line and in service if components are replaced
0x4077	Selective Catalytic Reductant Dosing Measuring Test	This routine carries out a leak and dosing quantity test by using inputs from the tool. If all inputs from the tool are zero, then the test is a visual inspection for leaks while under pressure, quantity tests are done by comparing the reductant amount delivered with the amount requested
0x4078	Selective Catalytic Reductant Emptying	This routine empties/purges the hydraulic lines of the DEF system which will be necessary in service to renew aged diesel exhaust fluid
0x4092	Test Reductant System Lines	This routine tests the integrity of the DEF lines at low pressure (1 - 2 bar)
0x4093	Test The Pump Emptying The High Pressure Reductant System	This routine is used to test the back flow pump which empties the DEF lines when the system is drained

0x4094	Test Dosing Valve Dynamic Behaviour	This routine tests the quantity of DEF delivered with the injector opened for a given period of time
0x4095	Test Measured Static Reductant Dosing Quantity	This routine tests the quantity of DEF delivered against an expected value
0x4096	Test Reductant Spray Pattern	This routine tests the pattern of DEF injected from the dosing module as there is often a build-up of residue around the actual injector.
0x4097	Prime Reductant System For First Fill	This routine is used to prime the reductant system following component replacement and includes a short pressure test to confirm the integrity of the pipework
0x409A	Reductant First Fill And Pressure Test	Selective reductant catalyst system end of line first fill and pressure test routine. This routine will prime and test the system

PARAMETER IDENTIFIER (PID)

PID	DESCRIPTION	ACTION
0x043C	Reductant Tank Temperature - Raw	- Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals - Signal is displayed in °C - Circuit reference I_T_RAULS Signal - Input to the engine control module PWM signal for DEF tank temperature
0x0441	Reductant Injector Duty Cycle	- Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals - Signal is displayed in % - Circuit reference O_T_RAMV_L - Output from the engine control module PWM signal to drive the DEF injector

0x05C1	Reductant Tank Fluid Level	- Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals - Signal is displayed in mm - Circuit reference I_T_RAULS Signal - Input to the engine control module PWM signal for DEF tank level
0x05C2	Reductant Tank Sensor Duty Cycles	- Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals - Signal is displayed in % - Signal value is read from internal memory location internal to engine control module over CAN communication bus
0x05CF	Reductant Purge Pump Duty Cycle - Commanded	- Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals - Signal is displayed in % - Circuit reference O_T_RAPM - Output from the engine control module PWM signal to drive the DEF pump
0x05D0	Reductant Pump Duty Cycle - Commanded	- Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals - Signal is displayed in % - Circuit reference O_T_RAPM - Output from the engine control module PWM signal to drive the DEF pump
0x05D1	Selective Catalyst Reductant Relay State - Commanded	- Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals - Signal is displayed as off/on - Circuit reference O_T_SCRMRLY - Output from the engine control module PWM signal to drive the DEF relay

0x05FB	Learnt Heater Current Threshold For Heater Run Dry Detection - Frozen Cavity Detection In Urea Tank	▪ Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals ▪ Signal is displayed in mA ▪ Signal value is read from internal memory location internal to engine control module over CAN communication bus
0x05FC	Total Mass Of Fluid Injected By Urea Dosing System	▪ Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals ▪ Signal is displayed in Kg ▪ Signal value is read from internal memory location internal to engine control module over CAN communication bus
0x05FD	Total Time Of Urea Dosing Valve Exceeding Upper Temperature Threshold	▪ Diagnosis may require using the manufacturer approved diagnostic system check datalogger signals ▪ Signal is displayed in s ▪ Signal value is read from internal memory location internal to engine control module over CAN communication bus
0xDD02	Main ECU Voltage Supply	▪ Diagnosis of these symptoms may require using the manufacturer approved diagnostic system check datalogger signals ▪ Signal is displayed in volts ▪ Signal value is read from internal memory location internal to engine control module over CAN communication bus
0xD940	Particulate Matter Sensor Meander Temperature – Raw	▪ Diagnosis of these symptoms may require using the manufacturer approved diagnostic system check datalogger signals ▪ Signal is displayed in °C ▪ Signal value is read from internal memory location internal to engine control module over CAN communication bus

DIESEL EXHAUST FLUID (DEF) LEVEL. 16MY EU_A 16MY NAS_A

APPLICATION	APPLICATION
16 MY Eu_A & Calibration part number	16 MY NAS_A & Calibration part number
Euro6b XE GX73-12K532-FAE/FBE/SAD/SBE	NAS LEVIII Range Rover V6D Mono 190 NAS GK52 -12K532-PRE
Japan XE GX73-12K532-FGA	NAS LEVIII Range Rover Sport V6D Mono 190 NAS GK52 -12K532-PRE
Euro6b new XF GX63-12K532-FAD/FBD/SAD/SBD	Not applicable
Japan new XF GX63-12K532-FGA	Not applicable
Euro6b new XF GX63-12K532-PCD	Not applicable
Euro6b XJ GW93-12K532-PCE	Not applicable
Euro6b Range Rover Mono 155 EU6 GK52 -12K532-PPE	Not applicable
Euro6b Range Rover Sport Mono 183 EU6 GK52 -12K532-PDE	Not applicable
Euro6b Range Rover HEV Twin 260 EU6 GPLA-12K532-HBF	Not applicable
Euro6b Range Rover Sport HEV Twin 260 EU6 GPLA-12K532-HBF	Not applicable
Euro6b RR Evoque GJ32-12K532-FBE/GAE/GTE/GBE/GLE/GCE/GDE	Not applicable
Euro6b LR Discovery Air Sus 188 EU6 GH22-12K532-PLE	Not applicable
Euro6b LR Discovery Sport Gk72-12K532-FBE/GAE/GTE/GBE/GND/GPD/GLE/GCE/GDE	Not applicable

The level of DEF in the tank is checked via the Instrument Cluster (IC) menu. When selected the message centre displays the approximate amount of

DEF remaining, providing an estimated km/miles figure remaining, prior to emptying the tank

The DEF tank is fitted with combined temperature/ultrasonic level sensors. Circuit continuity and plausibility of the signals received are checked within the ECM. An ultrasonic “no echo malfunction” is set if no valid echo from the ultrasonic sensor is detected during a predefined time. This typically happens when the ultrasonic signal of the sensor is blocked, or if the DEF tank is overfilled (the air pocket at the top of the DEF tank is no longer present). The vehicle message centre displays an advisory/warning system, which activates in low DEF conditions. If no corrective action is made, the engine will eventually not restart. The stages in the warning system are listed in the following tables

NOTE:

None of the listed stages will activate the MIL. The stages are automatically cleared if the DEF level is increased

LOW DEF LEVEL INSTRUMENT CLUSTER WARNING MESSAGE	DTC(S)	PURPOSE AND CAUSE	ACTION
▪ General warning message ▪ Diesel Exhaust Fluid Level Low ▪ MIL ▪ OFF	P2BAE-00	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 2400km / 1491 miles	Add diesel exhaust fluid
▪ Amber warning message ▪ Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ OFF	P2BAE-00	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 1500km / 932 miles	Add diesel exhaust fluid
▪ Amber warning message	P2BAF-00	▪ Low diesel exhaust	Add a

- No Engine Restarts Possible in 870km / 540 miles. Fill Diesel Exhaust Fluid Tank MIL - OFF	and P2BAE-00	fluid level Distance to NO ENGINE RESTART 870km / 540 miles	minimum 3 litres of diesel exhaust fluid
Amber warning message - No Engine Restarts Possible soon. Fill Diesel Exhaust Fluid Tank MIL - OFF	P2BAE-00 and P2BAF-00	- Low diesel exhaust fluid level Distance to NO ENGINE RESTART 160km / 100 miles	Add a minimum 3 litres of diesel exhaust fluid
Red warning message - No Engine Restarts Possible. Fill Diesel Exhaust Fluid Tank MIL - ON	P241D-00 and P2BAE-00 and P2BAF-00	- Low diesel exhaust fluid level Distance to NO ENGINE RESTART 0 km / 0 miles	Add a minimum 5 litres of diesel exhaust fluid

DIESEL EXHAUST FLUID (DEF) QUALITY. 16MY EU_A 16MY NAS_A

APPLICATION	APPLICATION
16 MY Eu_A & Calibration part number	16 MY NAS_A & Calibration part number
Euro6b XE GX73-12K532-FAE/FBE/SAD/SBE	NAS LEVIII Range Rover V6D Mono 190 NAS GK52 -12K532-PRE
Japan XE GX73-12K532-FGA	NAS LEVIII Range Rover Sport V6D Mono 190 NAS GK52 -12K532-PRE
Euro6b new XF GX63-12K532-FAD/FBD/SAD/SBD	Not applicable
Japan new XF GX63-12K532-FGA	Not applicable
Euro6b new XF GX63-12K532-PCD	Not applicable
Euro6b XJ GW93-12K532-PCE	Not applicable

Euro6b Range Rover Mono 155 EU6 GK52 -12K532-PPE	Not applicable
Euro6b Range Rover Sport Mono 183 EU6 GK52 -12K532-PDE	Not applicable
Euro6b Range Rover HEV Twin 260 EU6 GPLA- 12K532-HBF	Not applicable
Euro6b Range Rover Sport HEV Twin 260 EU6 GPLA-12K532-HBF	Not applicable
Euro6b RR Evoque GJ32-12K532- FBE/GAE/GTE/GBE/GLE/GCE/GDE	Not applicable
Euro6b LR Discovery Air Sus 188 EU6 GH22- 12K532-PLE	Not applicable
Euro6b LR Discovery Sport Gk72-12K532- FBE/GAE/GTE/GBE/GND/GPD/GLE/GCE/GDE	Not applicable

The DEF quality is monitored after refilling of the DEF tank. The monitoring begins when refilling is identified. In order to recognise a poor quality DEF, the OBD system uses the SCR catalytic converter NOx efficiencies, returned by the passive SCR catalytic converter monitoring, to build long and short term records of the average NOx conversion efficiencies. If the short term record becomes significantly different to the long term record, the malfunction is attributed to the quality of the DEF. The message centre warning system is activated when such a malfunction is detected. Simultaneously, the MIL is directly activated on the first occurrence of the malfunction detection

INCORRECT DEF QUALITY INSTRUMENT CLUSTER WARNING MESSAGE	DTC(S)	PURPOSE AND CAUSE	ACTION
- Amber warning message - Incorrect Diesel Exhaust Fluid Quality Detected - MIL - ON	P2BA9-00 and / or P2BA9-92 and P2BAE	- Incorrect diesel exhaust fluid quality - Distance to NO ENGINE RESTART	Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92

		870km / 540 miles	
▪ Amber warning message ▪ No Engine Restarts Possible in 830 km / 515 miles. Incorrect Diesel Exhaust Fluid Quality Detected ▪ MIL ▪ ON	P2BA9-00 and / or P2BA9-92 P2BAE-00 and P2BAF-00	▪ Incorrect diesel exhaust fluid quality ▪ Distance to NO ENGINE RESTART 830 km / 515 miles	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92
▪ Amber warning message ▪ No Engine Restarts Possible in 160 km / 100 miles. Incorrect Diesel Exhaust Fluid Quality Detected ▪ MIL ▪ ON	P2BA9-00 and / or P2BA9-92 P2BAE-00 and P2BAF-00	▪ Incorrect diesel exhaust fluid quality ▪ Distance to NO ENGINE RESTART 160 km / 100 miles	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92
▪ Red warning message ▪ No Engine Restarts Possible in 0 km / 0 miles. Incorrect Diesel Exhaust Fluid Quality Detected ▪ MIL ▪ ON	P2BA9-00 and / or P2BA9-92 P241D-00 and P2BAE-00 and P2BAF-00	▪ Incorrect diesel exhaust fluid quality ▪ Distance to NO ENGINE RESTART 0 km / 0 miles	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92 ▪ Using the manufacturer approved diagnostic system carry out Service Functions - Reset Selective Catalyst Reduction Start Inhibit ▪ NOTE: On restart 50 km / 31 miles allowed to follow repair procedure

Diesel Exhaust Fluid (DEF) Quality

Prior to the NOx sensor and the monitoring function becoming operational, the sensor must have reached its 'dew point'. This is the point where the moisture content in the exhaust gas has evaporated and can no longer damage the NOx sensor. In normal ambient conditions, this process typically takes around 6 to 10 minutes. If a malfunction is detected during the operation of the NOx sensor, a reversible fault is stored. If a malfunction is detected over two (NAS variants) or three (EU6 variants) consecutive drive cycles, a permanent fault is stored and the MIL is activated

The following monitoring is performed on the NOx sensors

- Signal plausibility and correction factor limits
- Circuit continuity between the sensor control unit and probe element
- Operating temperature reached after a defined heating period

Private CAN Bus - Monitoring

The following components are connected to the ECM via a private CAN bus. The ECM provides a gateway, for communication to the vehicle CAN bus

- Heater control unit
- NOx sensor control unit(s)

The following CAN communication monitoring applies to these components

- Signal time-out
- Signal check sum

The DEF injector, DEF pump, temperature and level sensors are hardwired to the ECM and do not involve CAN communication

DIESEL EXHAUST FLUID (DEF) SUPPLY
AND METERING SYSTEM MALFUNCTIONS.
16MY EU_A 16MY NAS_A

APPLICATION	APPLICATION
16 MY Eu_A & Calibration part number	16 MY NAS_A & Calibration part number
Euro6b XE GX73-12K532-FAE/FBE/SAD/SBE	NAS LEVIII Range Rover V6D Mono 190 NAS GK52 -12K532-PRE
Japan XE GX73-12K532-FGA	NAS LEVIII Range Rover Sport V6D Mono 190 NAS GK52 -12K532-PRE
Euro6b new XF GX63-12K532-FAD/FBD/SAD/SBD	Not applicable
Japan new XF GX63-12K532-FGA	Not applicable
Euro6b new XF GX63-12K532-PCD	Not applicable
Euro6b XJ GW93-12K532-PCE	Not applicable
Euro6b Range Rover Mono 155 EU6 GK52 -12K532-PPE	Not applicable
Euro6b Range Rover Sport Mono 183 EU6 GK52 -12K532-PDE	Not applicable
Euro6b Range Rover HEV Twin 260 EU6 GPLA-12K532-HBF	Not applicable
Euro6b Range Rover Sport HEV Twin 260 EU6 GPLA-12K532-HBF	Not applicable
Euro6b RR Evoque GJ32-12K532-FBE/GAE/GTE/GBE/GLE/GCE/GDE	Not applicable
Euro6b LR Discovery Air Sus 188 EU6 GH22-12K532-PLE	Not applicable
Euro6b LR Discovery Sport Gk72-12K532-FBE/GAE/GTE/GBE/GND/GPD/GLE/GCE/GDE	Not applicable

The OBD system monitors components and the functionality of the DEF supply and metering system. If a malfunction is detected a message centre warning is displayed. If no action is taken to rectify the warning malfunction, this will eventually lead to a 'No engine restart' situation

DEF SUPPLY AND METERING SYSTEM MALFUNCTION INSTRUMENT	DTC(S)	PURPOSE AND CAUSE	ACTION
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CLUSTER WARNING MESSAGE

▪ The MIL only will be displayed upon first DEF supply and metering system malfunction. At MIL only and 40 km / 25 miles then driven the instrument cluster warning message(s) will be seen ▪ MIL ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P21C7-13 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87	▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged
▪ Amber warning message ▪ No Engine Restarts Possible in 870 km / 540		▪ DEF supply and metering system malfunction DTC has	▪ Refer to the DTC helptext and carry out the repair actions for

miles. Diesel Exhaust Fluid Dosing System Malfunction ▪ MIL ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P2BAE-00 ▪ DTC(s) which may also be logged listed below ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P21C7-13 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87	been logged ▪ Distance to NO ENGINE RESTART 870km / 540 miles ▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	DTC(s) which are logged
▪ Amber warning message ▪ No Engine		▪ DEF supply and metering system	▪ Refer to the DTC helptext and carry out

Restarts Possible in 830 km / 515 miles. Diesel Exhaust Fluid Dosing System Malfunction ▪ MIL ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P2BAE-00 and P2BAF-00 ▪ DTC(s) which may also be logged listed below ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P21C7-13 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87	malfunction DTC has been logged ▪ Distance to NO ENGINE RESTART 830km / 515 miles ▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	the repair actions for DTC(s) which are logged
▪ Amber warning message		▪ DEF supply and metering	▪ Refer to the DTC helptext

- No Engine Restarts Possible in soon. Diesel Exhaust Fluid Dosing System Malfunction MIL - ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table - P2BAE-00 and P2BAF-00 DTC(s) which may also be logged listed below - P16A7-00 - P16A8-00 - P203C-00 - P203D-00 - P2047-13 - P208A-13 - P208B-00 - P20E8-84 - P20E8-85 - P20E9-84 - P20E9-85 - P21C7-13 - P249C-00 - P2BA7-7B - U029D-87 - U029E-87	system malfunction DTC has been logged Distance to NO ENGINE RESTART 160km / 100 miles - Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	and carry out the repair actions for DTC(s) which are logged
Red warning message		DEF supply and metering	Refer to the DTC helptext

▪ No Engine Restarts Possible ▪ MIL ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P241D-00 and P2BAE-00 and P2BAF-00 ▪ DTC(s) which may also be logged listed below ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P21C7-13 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87	system malfunction DTC has been logged ▪ Distance to NO ENGINE RESTART km / 0 miles ▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	and carry out the repair actions for DTC(s) which are logged ▪ Using the manufacturer approved diagnostic system carry out Service Functions - Reset Selective Catalyst Reduction Start Inhibit ▪ NOTE: On restart 50 km / 31 miles allowed to follow repair procedure
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Diesel Exhaust Fluid (DEF) Supply and Metering System Malfunctions

The following DEF supply and metering system malfunctions may trigger the message centre warning system

- Pressure build up in the supply line
- Metering/injection time to closed loop
- Over pressure in supply line
- Under pressure in supply line
- Injection valve open circuit
- SCR system relay open circuit
- Solenoid pump(s) open circuit
- Combined temperature/level sensor open circuit
- Missing communication with upstream NOx sensor (NAS variants only)
- Missing communication with downstream NOx sensor

DIESEL EXHAUST FLUID (DEF) LEVEL. 16MY EU_B 17MY EU_B

APPLICATION	APPLICATION
16 MY Eu_B and calibration part number	17 MY Eu_B and calibration part number
Euro6b Range Rover GPLA-12K532-NCE	Euro6b XE To be determined
Euro6b Range Rover Sport GPLA-12K532-NCE	Japan XE To be determined
Euro6b LR Discovery Air Sus 188 EU6 GH22-12K532-PLE	Euro6b new XF To be determined
Not applicable	Japan new XF To be determined
Not applicable	Euro6b new XF Not applicable
Not applicable	Euro6b F-Pace HK83-12K532-LAB/LBB/KCB
Not applicable	Japan F-Pace To be determined

Not applicable	Euro6b F-Pace HK83-12K532-PCE
Not applicable	Euro6b Range Rover HPLA-12K532-PGB/PHB/PJB
Not applicable	Euro6b Range Rover Sport HK62-12K532-PGB/PHB/PJB/PNB
Not applicable	Japan Range Rover HPLA-12K532-PSB
Not applicable	Japan Range Rover Sport HK62-12K532-PSB
Not applicable	Euro6b Range Rover HPLA-12K532-HNB
Not applicable	Euro6b Range Rover Sport HPLA-12K532-HNB
Not applicable	Euro6b Range Rover HK62-12K532-NNB
Not applicable	Euro6b Range Rover Sport HK62-12K532-NNB
Not applicable	Euro6b RR Evoque To be determined
Not applicable	Japan RR Evoque To be determined
Not applicable	Euro6b LR Discovery Sport To be determined
Not applicable	Japan LR Discovery Sport To be determined
Not applicable	NAS LEVIII LR Discovery Sport To be determined
Not applicable	Euro6b LR new Discovery HY32-12K532-PGB/PHB/PJB
Not applicable	Japan LR new Discovery HY32-12K532-PSB

The level of DEF in the tank is checked via the Instrument Cluster (IC) menu. When selected the message centre displays the approximate amount of DEF remaining, providing an estimated km/miles figure remaining, prior to emptying the tank

The DEF tank is fitted with combined temperature/ultrasonic level sensors. Circuit continuity and plausibility of the signals received are checked within the ECM. An ultrasonic "no echo malfunction" is set if no valid echo from the ultrasonic sensor is detected during a predefined time. This typically

happens when the ultrasonic signal of the sensor is blocked, or if the DEF tank is overfilled (the air pocket at the top of the DEF tank is no longer present). The vehicle message centre display's an advisory/warning system, which activates in low DEF conditions. If no corrective action is made, the engine will eventually not restart. The stages in the warning system are listed in the following tables

NOTE:

None of the listed stages will activate the MIL. The stages are automatically cleared if the DEF level is increased

LOW DEF LEVEL INSTRUMENT CLUSTER WARNING MESSAGE	DTC(S)	PURPOSE AND CAUSE	ACTION
▪ General warning message ▪ Diesel Exhaust Fluid Level Low ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ OFF	P2BAE-00	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 2400km / 1491 miles	Add diesel exhaust fluid
▪ Amber warning message ▪ Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ OFF	P2BAE-00 and P2BAE-02 (17MY on)	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 1500km / 932 miles	Add diesel exhaust fluid
▪ Amber warning message ▪ No Engine Restarts	P2BAF-00 and P2BAE-00 and P2BAE-02 (17MY on)	▪ Low diesel exhaust fluid level ▪ Distance	Add a minimum 3 litres of diesel

Possible in 870km / 540 miles. Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ ON		to NO ENGINE RESTART 870km / 540 miles	exhaust fluid
▪ Amber warning message ▪ No Engine Restarts Possible soon. Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ ON	P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on)	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 160km / 100 miles	Add a minimum 3 litres of diesel exhaust fluid
▪ Red warning message ▪ No Engine Restarts Possible. Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	P241D-00 and P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on) and P2BAF-04 (17MY on)	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 0 km / 0 miles	Add a minimum 5 litres of diesel exhaust fluid

DIESEL EXHAUST FLUID (DEF) QUALITY.
16MY EU_B 17MY EU_B

APPLICATION	APPLICATION

16 MY Eu_B and calibration part number	17 MY Eu_B and calibration part number
Euro6b Range Rover GPLA-12K532-NCE	Euro6b XE To be determined
Euro6b Range Rover Sport GPLA-12K532-NCE	Japan XE To be determined
Euro6b LR Discovery Air Sus 188 EU6 GH22-12K532-PLE	Euro6b new XF To be determined
Not applicable	Japan new XF To be determined
Not applicable	Euro6b new XF Not applicable
Not applicable	Euro6b F-Pace HK83-12K532-LAB/LBB/KCB
Not applicable	Japan F-Pace To be determined
Not applicable	Euro6b F-Pace HK83-12K532-PCE
Not applicable	Euro6b Range Rover HPLA-12K532-PGB/PHB/PJB
Not applicable	Euro6b Range Rover Sport HK62-12K532-PGB/PHB/PJB/PNB
Not applicable	Japan Range Rover HPLA-12K532-PSB
Not applicable	Japan Range Rover Sport HK62-12K532-PSB
Not applicable	Euro6b Range Rover HPLA-12K532-HNB
Not applicable	Euro6b Range Rover Sport HPLA-12K532-HNB
Not applicable	Euro6b Range Rover HK62-12K532-NNB
Not applicable	Euro6b Range Rover Sport HK62-12K532-NNB
Not applicable	Euro6b RR Evoque To be determined
Not applicable	Japan RR Evoque To be determined
Not applicable	Euro6b LR Discovery Sport To be determined
Not applicable	Japan LR Discovery Sport To be determined
Not applicable	NAS LEVIII LR Discovery Sport To be determined
Not applicable	Euro6b LR new Discovery HY32-12K532-PGB/PHB/PJB

Not applicable

Japan LR new Discovery HY32-12K532-PSB

INCORRECT DEF QUALITY INSTRUMENT CLUSTER WARNING MESSAGE	DTC(S)	PURPOSE AND CAUSE	ACTION
▪ Amber warning message ▪ Incorrect Diesel Exhaust Fluid Quality Detected ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	P2BA9-00 and / or P2BA9-67 and / or P2BA9-92 and P2BAE and P2BAE-02 (17MY on)	▪ Incorrect diesel exhaust fluid quality ▪ Distance to NO ENGINE RESTART 870km / 540 miles	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92
▪ Amber warning message ▪ No Engine Restarts Possible in 830 km / 515 miles. Incorrect Diesel Exhaust Fluid Quality Detected ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	P2BA9-00 and / or P2BA9-67 and / or P2BA9-92 P2BAF-00 and P2BAE-00 and P2BAE-02 (17MY on)	▪ Incorrect diesel exhaust fluid quality ▪ Distance to NO ENGINE RESTART 830 km / 515 miles	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92
▪ Amber	P2BA9-00 and / or P2BA9-67	▪ Incorrect	▪ Refer to the

warning message - No Engine Restarts Possible in 160 km / 100 miles. Incorrect Diesel Exhaust Fluid Quality Detected MIL - ON Reagent indicator (17MY on) - ON	and / or P2BA9-92 P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on)	diesel exhaust fluid quality Distance to NO ENGINE RESTART 160 km / 100 miles	DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92
Red warning message - No Engine Restarts Possible in 0 km / 0 miles. Incorrect Diesel Exhaust Fluid Quality Detected MIL - ON Reagent indicator (17MY on) - ON	P2BA9-00 and / or P2BA9-67 and / or P2BA9-92 P241D-00 and P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on) and P2BAF-04 (17MY on)	- Incorrect diesel exhaust fluid quality Distance to NO ENGINE RESTART 0 km / 0 miles	- Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-00 and / or P2BA9-92 - Using the manufacturer approved diagnostic system carry out Service Functions - Reset Selective Catalyst Reduction Start Inhibit NOTE: On restart 50 km / 31 miles allowed to follow repair procedure

Diesel Exhaust Fluid (DEF) Quality

Prior to the NOx sensor and the monitoring function becoming operational, the sensor must have reached its 'dew point'. This is the point where the moisture content in the exhaust gas has evaporated and can no longer damage the NOx sensor. In normal ambient conditions, this process typically takes around 6 to 10 minutes. If a malfunction is detected during the operation of the NOx sensor, a reversible fault is stored. If a malfunction is detected over two (NAS variants) or three (EU6 variants) consecutive drive cycles, a permanent fault is stored and the MIL is activated

The following monitoring is performed on the NOx sensors

- Signal plausibility and correction factor limits
- Circuit continuity between the sensor control unit and probe element
- Operating temperature reached after a defined heating period

Private CAN Bus - Monitoring

The following components are connected to the ECM via a private CAN bus. The ECM provides a gateway, for communication to the vehicle CAN bus

- Heater control unit
- NOx sensor control unit(s)

The following CAN communication monitoring applies to these components

- Signal time-out
- Signal check sum

The DEF injector, DEF pump, temperature and level sensors are hardwired to the ECM and do not involve CAN communication

DIESEL EXHAUST FLUID (DEF) SUPPLY AND METERING SYSTEM MALFUNCTIONS. 16MY EU_B 17MY EU_B

APPLICATION	APPLICATION
16 MY Eu_B and calibration part number	17 MY Eu_B and calibration part number
Euro6b Range Rover GPLA-12K532-NCE	Euro6b XE To be determined
Euro6b Range Rover Sport GPLA-12K532-NCE	Japan XE To be determined
Euro6b LR Discovery Air Sus 188 EU6 GH22-12K532-PLE	Euro6b new XF To be determined
Not applicable	Japan new XF To be determined
Not applicable	Euro6b new XF Not applicable
Not applicable	Euro6b F-Pace HK83-12K532-LAB/LBB/KCB
Not applicable	Japan F-Pace To be determined
Not applicable	Euro6b F-Pace HK83-12K532-PCE
Not applicable	Euro6b Range Rover HPLA-12K532-PGB/PHB/PJB
Not applicable	Euro6b Range Rover Sport HK62-12K532-PGB/PHB/PJB/PNB
Not applicable	Japan Range Rover HPLA-12K532-PSB
Not applicable	Japan Range Rover Sport HK62-12K532-PSB
Not applicable	Euro6b Range Rover HPLA-12K532-HNB
Not applicable	Euro6b Range Rover Sport HPLA-12K532-HNB
Not applicable	Euro6b Range Rover HK62-12K532-NNB
Not applicable	Euro6b Range Rover Sport HK62-12K532-NNB
Not applicable	Euro6b RR Evoque To be determined
Not applicable	Japan RR Evoque To be determined
Not applicable	Euro6b LR Discovery Sport To be determined

Not applicable	Japan LR Discovery Sport To be determined
Not applicable	NAS LEVIII LR Discovery Sport To be determined
Not applicable	Euro6b LR new Discovery HY32-12K532-PGB/PHB/PJB
Not applicable	Japan LR new Discovery HY32-12K532-PSB

The OBD system monitors components and the functionality of the DEF supply and metering system. If a malfunction is detected a message centre warning is displayed. If no action is taken to rectify the warning malfunction, this will eventually lead to a 'No engine restart' situation

DEF SUPPLY AND METERING SYSTEM MALFUNCTION INSTRUMENT CLUSTER WARNING MESSAGE	DTC(S)	PURPOSE AND CAUSE	ACTION
- The MIL only will be displayed upon first DEF supply and metering system malfunction. At MIL only and 40 km / 25 miles then driven the instrument cluster warning message(s) will be seen MIL - ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table - P05F2-72 - P16A7-00 - P16A8-00 - P203C-00 - P203D-00	Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged

	▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P208B-64 ▪ P208E-73 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P20FA-13 ▪ P21C7-13 ▪ P2470-16 ▪ P2471-17 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87		
▪ Amber warning message ▪ No Engine Restarts Possible in 870 km / 540 miles. Diesel Exhaust Fluid Dosing System Malfunction ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P2BAE-00 and P2BAE-02 (17MY on) ▪ DTC(s) which may also be logged listed below	▪ DEF supply and metering system malfunction DTC has been logged ▪ Distance to NO ENGINE RESTART 870km / 540 miles ▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged

	▪ P05F2-72 ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P208B-64 ▪ P208E-73 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P20FA-13 ▪ P21C7-13 ▪ P2470-16 ▪ P2471-17 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87		
▪ Amber warning message ▪ No Engine Restarts Possible in 830 km / 515 miles. Diesel Exhaust Fluid Dosing System Malfunction ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table	▪ DEF supply and metering system malfunction DTC has been logged ▪ Distance to NO ENGINE RESTART 830km / 515 miles ▪ Refer to the DTC helptext for purpose and cause. Carry out the	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged

	▪ P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 ▪ DTC(s) which may also be logged listed below ▪ P05F2-72 ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P208B-64 ▪ P208E-73 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P20FA-13 ▪ P21C7-13 ▪ P2470-16 ▪ P2471-17 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87	repair actions for DTC(s) which are logged	
▪ Amber warning message ▪ No Engine Restarts Possible in soon. Diesel Exhaust Fluid Dosing System Malfunction ▪ MIL ▪ ON		▪ DEF supply and metering system malfunction DTC has been logged ▪ Distance to NO ENGINE RESTART 160km /	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged

■ **Reagent indicator
(17MY on)**

- ON

NOTE:

The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table

- P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on)

■ **DTC(s) which may also be logged listed below**

- P05F2-72
- P16A7-00
- P16A8-00
- P203C-00
- P203D-00
- P2047-13
- P208A-13
- P208B-00
- P208B-64
- P208E-73
- P20E8-84
- P20E8-85
- P20E9-84
- P20E9-85
- P20FA-13
- P21C7-13
- P2470-16

100
miles

- Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged

	▪ P2471-17 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87		
▪ Red warning message ▪ No Engine Restarts Possible ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P241D-00 and P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on) and P2BAF-04 (17MY on) ▪ DTC(s) which may also be logged listed below ▪ P05F2-72 ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P208B-64	▪ DEF supply and metering system malfunction DTC has been logged ▪ Distance to NO ENGINE RESTART km / 0 miles ▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged ▪ Using the manufacturer approved diagnostic system carry out Service Functions - Reset Selective Catalyst Reduction Start Inhibit ▪ NOTE: On restart 50 km / 31 miles allowed to follow repair procedure

	▪ P208E-73 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P20FA-13 ▪ P21C7-13 ▪ P2470-16 ▪ P2471-17 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87		
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Diesel Exhaust Fluid (DEF) Supply and Metering System Malfunctions

The following DEF supply and metering system malfunctions may trigger the message centre warning system

- Pressure build up in the supply line
 - Metering/injection time to closed loop
 - Over pressure in supply line
 - Under pressure in supply line
 - Injection valve open circuit
 - SCR system relay open circuit
 - Solenoid pump(s) open circuit
 - Combined temperature/level sensor open circuit
 - Missing communication with upstream NOx sensor (NAS variants only)
 - Missing communication with downstream NOx sensor
-

DIESEL EXHAUST FLUID (DEF) LEVEL. 16MY NAS_B - 17MY NAS_B

APPLICATION	APPLICATION
16 MY NAS_B and calibration part number	17 MY NAS_B and calibration part number
NAS LEVIII Range Rover GK52 -12K532-PRF	NAS LEVIII XE To be determined
NAS LEVIII Range Rover Sport GK62-12K532-PRF	NAS LEVIII new XF To be determined
Not applicable	NAS LEVIII F-Pace To be determined
Not applicable	NAS LEVIII Range Rover HPLA-12K532-PKB
Not applicable	NAS LEVIII Range Rover Sport HK62-12K532-PKB
Not applicable	NAS LEVIII RR Evoque To be determined
Not applicable	NAS LEVIII LR new Discovery HY32-12K532-PKB

The level of DEF in the tank is checked via the Instrument Cluster (IC) menu. When selected the message centre displays the approximate amount of DEF remaining, providing an estimated km/miles figure remaining, prior to emptying the tank

The DEF tank is fitted with combined temperature/ultrasonic level sensors. Circuit continuity and plausibility of the signals received are checked within the ECM. An ultrasonic "no echo malfunction" is set if no valid echo from the ultrasonic sensor is detected during a predefined time. This typically happens when the ultrasonic signal of the sensor is blocked, or if the DEF tank is overfilled (the air pocket at the top of the DEF tank is no longer present). The vehicle message centre display's an advisory/warning system, which activates in low DEF conditions. If no corrective action is made, the engine will eventually not restart. The stages in the warning system are listed in the following tables

NOTE:

None of the listed stages will activate the MIL. The stages are automatically cleared if the DEF level is increased

LOW DEF LEVEL INSTRUMENT CLUSTER WARNING MESSAGE	DTC(S)	PURPOSE AND CAUSE	ACTION
▪ General warning message ▪ Diesel Exhaust Fluid Level Low ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ OFF	P2BAE-00	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 1500 miles / 2415km	Add diesel exhaust fluid
▪ Amber warning message ▪ Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ OFF	P2BAE-00 and P2BAE-02 (17MY on)	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 1000 miles / 1610 km	Add diesel exhaust fluid
▪ Amber warning message ▪ No Engine Restarts Possible in 500 miles / 806 km. Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on)	P2BAF-00 and P2BAE-00 and P2BAE-02 (17MY on)	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 500 miles / 806 km	Add a minimum 3 litres of diesel exhaust fluid

▪ ON			
▪ Red warning message ▪ No Engine Restarts Possible. Fill Diesel Exhaust Fluid Tank ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ ON	P241D-00 and P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on) and P2BAF-04 (17MY on)	▪ Low diesel exhaust fluid level ▪ Distance to NO ENGINE RESTART 0 miles / 0 km	Add a minimum 5 litres of diesel exhaust fluid

DIESEL EXHAUST FLUID (DEF) QUALITY. 16MY NAS_B - 17MY NAS_B

APPLICATION	APPLICATION
16 MY NAS_B and calibration part number	17 MY NAS_B and calibration part number
NAS LEVIII Range Rover GK52 -12K532-PRF	NAS LEVIII XE To be determined
NAS LEVIII Range Rover Sport GK62-12K532-PRF	NAS LEVIII new XF To be determined
Not applicable	NAS LEVIII F-Pace To be determined
Not applicable	NAS LEVIII Range Rover HPLA-12K532-PKB
Not applicable	NAS LEVIII Range Rover Sport HK62-12K532-PKB
Not applicable	NAS LEVIII RR Evoque To be determined
Not applicable	NAS LEVIII LR new Discovery HY32-12K532-PKB

INCORRECT DEF QUALITY INSTRUMENT CLUSTER WARNING	DTC(S)	PURPOSE AND CAUSE	ACTION
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MESSAGE

▪ Amber warning message ▪ No Engine Restarts Possible in 250 miles / 404 km. Incorrect Diesel Exhaust Fluid Quality Detected ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ ON	P2BA9-67 and / or P2BA9-92 P2BAF-00 and P2BAE-00 and P2BAE-02 (17MY on)	▪ Incorrect diesel exhaust fluid quality ▪ Distance to NO ENGINE RESTART 250 miles / 404 km	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-67 and / or P2BA9-92
▪ Red warning message ▪ No Engine Restarts Possible in 0 miles / 0 km. Incorrect Diesel Exhaust Fluid Quality Detected ▪ MIL ▪ OFF ▪ Reagent indicator (17MY on) ▪ ON	P2BA9-67 and / or P2BA9-92 P241D-00 and P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on) and P2BAF-04 (17MY on)	▪ Incorrect diesel exhaust fluid quality ▪ Distance to NO ENGINE RESTART 0 miles / 0 km	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) P2BA9-67 and / or P2BA9-92 ▪ Using the manufacturer approved diagnostic system carry out Service Functions - Reset Selective Catalyst Reduction Start Inhibit ▪ NOTE: On restart 50 km / 31 miles allowed to follow repair procedure

Diesel Exhaust Fluid (DEF) Quality

Prior to the NOx sensor and the monitoring function becoming operational, the sensor must have reached its 'dew point'. This is the point where the moisture content in the exhaust gas has evaporated and can no longer damage the NOx sensor. In normal ambient conditions, this process typically takes around 6 to 10 minutes. If a malfunction is detected during the operation of the NOx sensor, a reversible fault is stored. If a malfunction is detected over two (NAS variants) or three (EU6 variants) consecutive drive cycles, a permanent fault is stored and the MIL is activated

The following monitoring is performed on the NOx sensors

- Signal plausibility and correction factor limits
- Circuit continuity between the sensor control unit and probe element
- Operating temperature reached after a defined heating period

Private CAN Bus - Monitoring

The following components are connected to the ECM via a private CAN bus. The ECM provides a gateway, for communication to the vehicle CAN bus

- Heater control unit
- NOx sensor control unit(s)

The following CAN communication monitoring applies to these components

- Signal time-out
- Signal check sum

The DEF injector, DEF pump, temperature and level sensors are hardwired to the ECM and do not involve CAN communication

DIESEL EXHAUST FLUID (DEF) SUPPLY AND METERING SYSTEM MALFUNCTIONS. 16MY NAS_B - 17MY NAS_B

APPLICATION	APPLICATION
16 MY NAS_B and calibration part number	17 MY NAS_B and calibration part number
NAS LEVIII Range Rover GK52 -12K532-PRF	NAS LEVIII XE To be determined
NAS LEVIII Range Rover Sport GK62-12K532-PRF	NAS LEVIII new XF To be determined
Not applicable	NAS LEVIII F-Pace To be determined
Not applicable	NAS LEVIII Range Rover HPLA-12K532-PKB
Not applicable	NAS LEVIII Range Rover Sport HK62-12K532-PKB
Not applicable	NAS LEVIII RR Evoque To be determined
Not applicable	NAS LEVIII LR new Discovery HY32-12K532-PKB

The OBD system monitors components and the functionality of the DEF supply and metering system. If a malfunction is detected a message centre warning is displayed. If no action is taken to rectify the warning malfunction, this will eventually lead to a 'No engine restart' situation

DEF SUPPLY AND METERING SYSTEM MALFUNCTION INSTRUMENT CLUSTER WARNING MESSAGE	DTC(S)	PURPOSE AND CAUSE	ACTION
▪ Amber warning message ▪ No Engine Restarts Possible in 250 miles / 403 km.		▪ DEF supply and metering system malfunction DTC has been logged ▪ Distance to NO	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged

Diesel Exhaust Fluid Dosing System Malfunction ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 ▪ DTC(s) which may also be logged listed below ▪ P05F2-72 ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P208B-64 ▪ P208E-73 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P20FA-13 ▪ P21C7-13 ▪ P2470-16 ▪ P2471-17 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87	ENGINE RESTART 250 miles / 403 km ▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged
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	▪ U029E-87		
▪ Red warning message ▪ No Engine Restarts Possible ▪ MIL ▪ ON ▪ Reagent indicator (17MY on) ▪ ON	NOTE: The following DTC(s) are related to the DEF SUPPLY AND METERING SYSTEM and will be responsible for the Instrument cluster warning message(s) seen in this table ▪ P241D-00 and P2BAE-00 and P2BAE-02 (17MY on) and P2BAF-00 and P2BAF-02 (17MY on) and P2BAF-04 (17MY on) ▪ DTC(s) which may also be logged listed below ▪ P05F2-72 ▪ P16A7-00 ▪ P16A8-00 ▪ P203C-00 ▪ P203D-00 ▪ P2047-13 ▪ P208A-13 ▪ P208B-00 ▪ P208B-64 ▪ P208E-73 ▪ P20E8-84 ▪ P20E8-85 ▪ P20E9-84 ▪ P20E9-85 ▪ P20FA-13 ▪ P21C7-13 ▪ P2470-16	▪ DEF supply and metering system malfunction DTC has been logged ▪ Distance to NO ENGINE RESTART 0 miles / 0 km ▪ Refer to the DTC helptext for purpose and cause. Carry out the repair actions for DTC(s) which are logged	▪ Refer to the DTC helptext and carry out the repair actions for DTC(s) which are logged ▪ Using the manufacturer approved diagnostic system carry out Service Functions - Reset Selective Catalyst Reduction Start Inhibit ▪ NOTE: On restart 50 km / 31 miles allowed to follow repair procedure

	▪ P2471-17 ▪ P249C-00 ▪ P2BA7-7B ▪ U029D-87 ▪ U029E-87		
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Diesel Exhaust Fluid (DEF) Supply and Metering System Malfunctions

The following DEF supply and metering system malfunctions may trigger the message centre warning system

- Pressure build up in the supply line
- Metering/injection time to closed loop
- Over pressure in supply line
- Under pressure in supply line
- Injection valve open circuit
- SCR system relay open circuit
- Solenoid pump(s) open circuit
- Combined temperature/level sensor open circuit
- Missing communication with upstream NOx sensor (NAS variants only)
- Missing communication with downstream NOx sensor

Q A N D A

QUESTION	ANSWER
Is the inducement system only triggered by low levels of DEF?	NO. The inducement system ensures correct operation of the SCR/DEF system. It monitors (1) DEF level (2) DEF quality (3) Dosing hardware. A low level (1) OR a malfunction detected in area (2) or (3) will trigger the inducement system. The warning messages states the malfunction.
Is the number	NO. The engine can be stopped / started indefinitely until the message

of engine restarts limited while the vehicle is driven with the inducement system on?	"No Engine Restart Possible" shows.
Is there sufficient warning before the engine does NOT restart?	YES. The mileage countdown shown by the inducement system is guaranteed mileage before the no restart condition. The vehicle can be driven on the final "No Engine Restart Possible" warning until the engine is stopped by the driver. An engine stall is NOT counted as an engine stop. NOTE: Though an ECO stop is not counted as an engine stop; ECO stop functionality is NO longer available when the final "No Engine Restart Possible" warning shows.
Can the inducement system "heal" if the malfunction no longer exists?	YES. The inducement system will heal automatically - warnings will disappear - if the malfunction no longer exists. If the malfunction was to re-occur within 40 hours of operation, the inducement system will resume at the mileage it was left. This is to prevent long term tampering with the system.
Are the P2BAE / P2BAF DTC(s) to represent a malfunction?	NO. The P2BAE / P2BAF DTC(s) only help to understand at what level of inducement the vehicle currently is.
Is the low DEF level a malfunction?	NO. This is why no other DTC(s) than P2BAE / P2BAF - to represent the level of inducement the vehicle is in - are visible when low DEF occurs. The DEF tank shall be topped up with necessary volumes.
Is there a minimum top up volume when DEF level is low?	YES. More than 3 litres DEF or 2 "KRUSE" bottles or 1 US gallon bottle are necessary to activate top up detection. This also applies when the vehicle does NOT restart because of low DEF level.

DIESEL EXHAUST FLUID

DTC	DESCRIPTION	PURPOSE AND CAUSE	ACTION
P05ED-16	Reductant Heater Control Module Supply Voltage -		

Circuit voltage
below threshold

NOTE:

To monitor
diesel exhaust
fluid heater
control unit.
Diesel exhaust
fluid heater
control unit
under voltage

- Purpose of the DTC
 - To monitor diesel exhaust fluid heater control unit
- Electrical Cause
 - Yes
- Mechanical Cause
 - No
- Control Module Cavity
 - Circuit reference B_D_CANH3
 - Circuit reference B_D_CANL3
- Monitor Description
 - Diesel exhaust fluid heater control unit under voltage
- Prioritised List of Possible Causes
 - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground
 - Diesel exhaust fluid heater control unit power and ground circuits

NOTE:

This DTC when logged with an intermittent status on its own is advisory only and no further action should be taken

- Using the manufacturer approved diagnostic system, check for related DTCs
- If this DTC exists with any other diesel exhaust fluid DTCs, follow the advise given for that DTC
- Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
- Vehicle Conditions to enable DTC Logging Strategy
 - Engine running
- Prioritised Checks to Perform
 - Refer to the electrical circuit diagrams and check diesel exhaust fluid heater control unit power and ground circuits for open circuit
 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Check and install a new diesel exhaust fluid heater control unit as required
 - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		open circuit ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P05ED-17	Reductant Heater Control Module Supply Voltage - Circuit voltage above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference B_D_CANH3 ▪ Circuit reference B_D_CANL3 ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit over voltage ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check diesel exhaust fluid heater control unit power and ground circuits for short circuit to ground, circuit short circuit to power ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, circuit short circuit to power ▪ inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the manufacturer approved diagnostic system

		specified range but not necessarily a short circuit to power ▪ Diesel exhaust fluid heater control unit power or ground circuit short circuit to ground, circuit short circuit to power ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P05F1-73	Reductant Pump Supply Voltage Circuit Stuck On - Actuator stuck closed	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid relay stuck	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to

		closed ▪ Prioritised List of Possible Causes ▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P05F2-72	Reductant Pump Supply Voltage Circuit Stuck Off - Actuator stuck open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance

		▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid relay stuck open ▪ Prioritised List of Possible Causes ▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to close some piece of equipment ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P05F8-13	Reductant Heater Control Module Performance - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater

		■ Mechanical Cause ■ No ■ Control Module Cavity ■ Circuit reference B_D_CANH3 ■ Circuit reference B_D_CANL3 ■ Monitor Description ■ Diesel exhaust fluid heater control unit missing voltage ■ Prioritised List of Possible Causes ■ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ■ Diesel exhaust fluid heater control unit power or ground circuit open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Diesel exhaust fluid heater control unit failure	control unit circuit for open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install a new diesel exhaust fluid heater control unit as required ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P05F8-4B	Reductant Heater Control Module Performance - Over	■ Purpose of the DTC ■ To monitor diesel exhaust	■ Vehicle Conditions to enable DTC Logging Strategy ■ Engine running

	temperature	fluid heater control unit - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference B_D_CANH3 - Circuit reference B_D_CANL3 - Monitor Description - Diesel exhaust fluid heater control unit over temperature - Prioritised List of Possible Causes - The engine control module detected an internal temperature above the expected range - Diesel exhaust fluid heater control unit circuit short circuit to power - Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid heater control unit failure	- Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to power - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid heater control unit module as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P05F8-66	Reductant Heater Control Module Performance - Signal has too many transitions /	- Purpose of the DTC - To monitor diesel exhaust fluid heater control unit	- Vehicle Conditions to enable DTC Logging Strategy - Engine running - Prioritised Checks to Perform

events

- Electrical Cause
 - Yes
- Mechanical Cause
 - No
- Control Module Cavity
 - Circuit reference B_D_CANH3
 - Circuit reference B_D_CANL3
- Monitor Description
 - Diesel exhaust fluid heater control unit non volatile memory error
- Prioritised List of Possible Causes
 - The engine control module monitored a parameter over time within specified limits and detected more than the expected number of transitions
 - Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance
 - Connector is disconnected, connector pin is backed out, connector pin corrosion
 - Diesel exhaust fluid heater control unit failure
- Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power, open circuit, high resistance
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- Check and install a new diesel exhaust fluid heater control unit as required
- Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P16A3-00	SCR System - EOL - Failure During Emptying Test - No sub type information	▪ Purpose of the DTC ▪ Selective catalytic reduction dosing system prime routine failed during the empty phase ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Selective catalytic reduction dosing system prime routine failed during the empty phase ▪ Prioritised List of Possible Causes ▪ Selective catalytic reduction priming routine did not complete successfully ▪ DEF back flow pump did not empty the pressure line correctly ▪ DEF pressure line leakage ▪ Routine aborted at ignition off	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ 0x043C Reductant Tank Temperature - Raw ▪ Check that the DEF tank has at least 25mm of fluid (0x05C1) ▪ Check the DEF tank temperature is within -5 to + 70 °C (0x043C) ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check the DEF pressure line for leakage ▪ Using the manufacturer approved diagnostic system carry out routine 0x409A - Reductant First Fill And Pressure Test. Check that the ignition stays on throughout the whole duration of the routine ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P16A4-00	SCR System - EOL - Failure During First Fill Test - No sub type information	▪ Purpose of the DTC ▪ Selective catalytic reduction dosing	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on

		system prime routine failed during the first fill phase ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Selective catalytic reduction dosing system prime routine failed during the first fill phase ▪ Prioritised List of Possible Causes ▪ Selective catalytic reduction priming routine did not complete successfully ▪ Pressure and DEF back flow pump not activated correctly ▪ Routine aborted at ignition off	▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ 0x043C Reductant Tank Temperature - Raw ▪ Check that the DEF tank has at least 25mm of fluid (0x05C1) ▪ Check the DEF tank temperature is within -5 to + 70 °C (0x043C) ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check the DEF pressure line for leakage ▪ Using the manufacturer approved diagnostic system carry out routine 0x409A - Reductant First Fill And Pressure Test. Check that the ignition stays on throughout the whole duration of the routine ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P16A5-00	SCR System - EOL - Failure During Pressure Test - No sub type information	▪ Purpose of the DTC ▪ Selective catalytic reduction dosing system prime routine failed during the pressure preparation	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals

		phase - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - Selective catalytic reduction dosing system prime routine failed during the pressure preparation phase - Prioritised List of Possible Causes - Selective catalytic reduction priming routine did not complete successfully - Pressure pump is not pressurizing the line correctly - DEF pressure line leakage - Routine aborted at ignition off	0x05C1 Reductant Tank Fluid Level 0x043C Reductant Tank Temperature - Raw - Check that the DEF tank has at least 25mm of fluid (0x05C1) - Check the DEF tank temperature is within -5 to + 70 °C (0x043C) - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Check the DEF pressure line for leakage - Using the manufacturer approved diagnostic system carry out routine 0x409A - Reductant First Fill And Pressure Test. Check that the ignition stays on throughout the whole duration of the routine - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P16A6-00	SCR System - EOL - Unfinished Tests - No sub type information	- Purpose of the DTC - Selective catalytic reduction dosing system prime routine aborted during cycle - Electrical Cause - No - Mechanical Cause - No - Control Module	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on - Prioritised Checks to Perform - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x05C1 Reductant Tank Fluid Level 0x043C Reductant Tank

		- Cavity - Not applicable - Monitor Description - Selective catalytic reduction dosing system prime routine aborted during cycle - Prioritised List of Possible Causes - Selective catalytic reduction priming routine did not complete successfully - The routine has been aborted by the user - Routine aborted at ignition off	Temperature - Raw - Check that the DEF tank has at least 25mm of fluid (0x05C1) - Check the DEF tank temperature is within -5 to + 70 °C (0x043C) - Using the manufacturer approved diagnostic system carry out routine 0x409A - Reductant First Fill And Pressure Test. Check that the ignition stays on throughout the whole duration of the routine - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P16A7-00	SCR System - Tampering At Connector 1 - No sub type information	- Purpose of the DTC - Diesel exhaust fluid tank module connector monitoring (pump) - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid tank module (pump) connector is removed - Prioritised List of Possible Causes - Diesel exhaust	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start - Prioritised Checks to Perform - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Vehicle Conditions to enable DTC Logging Strategy

		fluid tank module connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Engine running ▪ On completion of the previous repair actions the following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P16A8-00	SCR System - Tampering At Connector 2 - No sub type information	▪ Purpose of the DTC ▪ Diesel exhaust fluid tank module connector monitoring (heater/level) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable	

- Monitor Description
 - Diesel exhaust fluid tank module (heater/level) connector is removed
- Prioritised List of Possible Causes
 - Diesel exhaust fluid tank module connector is disconnected, connector pin is backed out, connector pin corrosion

NOTE:

A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start

- Prioritised Checks to Perform
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- Vehicle Conditions to enable DTC Logging Strategy
 - Engine running
 - On completion of the previous repair actions the following MUST be carried out
 - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
 - NOTE: The engine MIL is OFF (when engine ON)
 - Start engine
 - The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle
 - NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required

			▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P202A-13	Reductant Tank Heater Control Circuit/Open - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater for open circuit, high resistance ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid heater control unit circuit open circuit, high resistance ▪ Connector is disconnected,	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P202B-11	Reductant Tank Heater Control Circuit Low - Circuit short to ground	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater for short circuit to ground ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater short circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		- Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid heater control unit failure	
P202B-13	Reductant Tank Heater Control Circuit Low - Circuit open	- Purpose of the DTC - To monitor diesel exhaust fluid heater for open circuit, high resistance - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid heater open circuit, high resistance - Prioritised List of Possible Causes - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Diesel exhaust fluid heater control unit circuit open circuit, high resistance	- Vehicle Conditions to enable DTC Logging Strategy - DEF heater on - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance - Refer to the electrical circuit diagrams and check the CAN communication diesel exhaust fluid tank module - DEF heater controller circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid heater control unit as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ CAN communication diesel exhaust fluid tank module - DEF heater controller circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P202C-12	Reductant Tank Heater Control Circuit High - Circuit short to battery	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater for short circuit to power ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater short circuit to power ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		vehicle power measurement when another value was expected ▪ Diesel exhaust fluid heater control unit circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P202C-15	Reductant Tank Heater Control Circuit High - Circuit short to battery or open	▪ Purpose of the DTC ▪ To monitor DEF tank heater element ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ DEF tank heater element short circuit ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a power measurement or open circuit for a period longer than expected or has detected a power	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit short circuit to ground short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		measurement or open circuit when another value was expected ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P202E-4B	Reductant Injection Valve Circuit Range/Performance (Bank 1 Unit 1) - Over temperature	▪ Purpose of the DTC ▪ To monitor DEF injector for over temperature ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference RAMV_L ▪ Circuit reference RAMV_H ▪ Monitor Description ▪ DEF injector over temperature ▪ Prioritised List of Possible Causes ▪ The engine control module detected an internal temperature	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check diesel exhaust fluid injector for overheating condition ▪ Check and install a new diesel exhaust fluid injector as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		above the expected range ▪ Diesel exhaust fluid injector circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid injector overheated ▪ Diesel exhaust fluid injector failure	
P203B-02	Reductant Level Sensor "A" Circuit Range/Performance - General signal failure	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor (level) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Diesel exhaust fluid tank module level reading not possible ▪ Prioritised List of Possible Causes ▪ Overfilled DEF tank - Refer to D & T section 309-00C: Exhaust system and refer to functionality of level sensor (possibility that	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Tank unfrozen, drive for 10 Km ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x043C Reductant Tank Temperature - Raw ▪ Check the DEF tank temperature is within -5 to + 70 °C ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check, drain and correct diesel exhaust fluid level ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins

		the sensor reads 0 if over filled) - Customer has over filled the DEF tank - Diesel exhaust fluid tank module communication error - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	for damage and/or corrosion - Contact dealer technical support - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P203B-16	Reductant Level Sensor "A" Circuit Range/Performance - Circuit voltage below threshold	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (level) - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference I_T_RAULS Signal - Monitor Description - Diesel exhaust fluid tank module level signal range check too low - Prioritised List of	- Vehicle Conditions to enable DTC Logging Strategy - Engine running - Prioritised Checks to Perform - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x043C Reductant Tank Temperature - Raw - Check the DEF tank temperature is within -5 to + 70 °C 0x05C1 Reductant Tank Fluid Level - Check, drain and correct diesel exhaust fluid level - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short

		Possible Causes - Overfilled DEF tank - Refer to D & T section 309-00C: Exhaust system and refer to functionality of level sensor (possibility that the sensor reads 0 if over filled) - Customer has over filled the DEF tank - The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P203B-17	Reductant Level Sensor "A" Circuit Range/Performance - Circuit voltage above threshold	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (level) - Electrical Cause - Yes - Mechanical Cause - No - Control Module	- Vehicle Conditions to enable DTC Logging Strategy - Engine running - Prioritised Checks to Perform - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x043C Reductant Tank Temperature - Raw

		Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Diesel exhaust fluid tank module level signal range check too high ▪ Prioritised List of Possible Causes ▪ Overfilled DEF tank - Refer to D & T section 309-00C: Exhaust system and refer to functionality of level sensor (possibility that the sensor reads 0 if over filled) ▪ Customer has over filled the DEF tank ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check the DEF tank temperature is within -5 to + 70 °C ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check, drain and correct diesel exhaust fluid level ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	

P203B-31	Reductant Level Sensor "A" Circuit Range/Performance - No signal	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor (level) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ DEF level value not available ▪ Prioritised List of Possible Causes ▪ Overfilled DEF tank - Refer to D & T section 309-00C: Exhaust system and refer to functionality of level sensor (possibility that the sensor reads 0 if over filled) ▪ Customer has over filled the DEF tank ▪ The engine control module does not detect a signal which ought to be present ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected,	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF tank unfrozen ▪ Drive 10 Km ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x043C Reductant Tank Temperature - Raw ▪ Check the DEF tank temperature is within -5 to + 70 °C ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check, drain and correct diesel exhaust fluid level ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
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		connector pin is backed out, connector pin corrosion	
P203C-00	Reductant Level Sensor "A" Circuit Low - No sub type information	■ Purpose of the DTC ■ Monitoring of the diesel exhaust fluid level sensor (level) ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ No ■ Control Module Cavity ■ Circuit reference I_T_RAULS Signal ■ Monitor Description ■ Diesel exhaust fluid tank module level sensor low level line error ■ Prioritised List of Possible Causes ■ Overfilled DEF tank - Refer to D & T section 309-00C: Exhaust system and refer to functionality of level sensor (possibility that the sensor reads 0 if over filled) ■ Customer has over filled the DEF tank ■ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start ■ Prioritised Checks to Perform ■ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ■ 0x05C1 Reductant Tank Fluid Level ■ Check, drain and correct diesel exhaust fluid level ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Contact dealer technical support ■ Vehicle Conditions to enable DTC Logging Strategy ■ Engine running

		Connector is disconnected, connector pin is backed out, connector pin corrosion	- On completion of the previous repair actions the following MUST be carried out - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest - NOTE: The engine MIL is OFF (when engine ON) - Start engine - The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle - NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault storage from P2BAE / P2BAF
P203D-00	Reductant Level Sensor "A" Circuit High - No sub type information	- Purpose of the DTC - Monitoring of the diesel exhaust fluid level sensor (level) - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference I_T_RAULS Signal	

- Monitor Description
 - Diesel exhaust fluid tank module level sensor high level line error
- Prioritised List of Possible Causes
 - Overfilled DEF tank - Refer to D & T section 309-00C: Exhaust system and refer to functionality of level sensor (possibility that the sensor reads 0 if over filled)
 - Customer has over filled the DEF tank
 - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance
 - Connector is disconnected, connector pin is backed out, connector pin corrosion

NOTE:

A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start

- Prioritised Checks to Perform
 - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
 - 0x05C1 Reductant Tank Fluid Level
 - Check, drain and correct diesel exhaust fluid level
 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Contact dealer technical support
- Vehicle Conditions to enable DTC Logging Strategy
 - Engine running
 - On completion of the previous repair actions the following MUST be carried out
 - Using the manufacturer

			approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P2047-13	Reductant Injection Valve Circuit / Open (Bank 1 Unit 1) - Circuit open	▪ Purpose of the DTC ▪ To monitor DEF injector circuit for open circuit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAMV_L ▪ Circuit reference RAMV_H ▪ Monitor Description ▪ DEF injector circuit open circuit, high resistance ▪ Prioritised List of	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for open circuit, high

		Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid injector circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid injector failure	resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid injector as required ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ On completion of the previous repair actions the following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P2048-11	Reductant Injection Valve Circuit Low (Bank 1 Unit 1) - Circuit short to	▪ Purpose of the DTC ▪ To monitor DEF injector circuit for short circuit	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform

	ground	to ground ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAMV_L ▪ Circuit reference RAMV_H ▪ Monitor Description ▪ DEF injector circuit for short circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid injector circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid injector failure	▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid injector as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2049-12	Reductant Injection Valve Circuit High	▪ Purpose of the DTC ▪ To monitor DEF	▪ Vehicle Conditions to enable DTC Logging Strategy

(Bank 1 Unit 1) -
Circuit short to
battery

injector circuit
for short circuit
to power

- Electrical Cause
 - Yes
- Mechanical Cause
 - No
- Control Module Cavity
 - Circuit reference RAMV_L
 - Circuit reference RAMV_H
- Monitor Description
 - DEF injector circuit for short circuit to ground
- Prioritised List of Possible Causes
 - The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected
 - Diesel exhaust fluid injector circuit short circuit to power
 - Connector is disconnected, connector pin is backed out, connector pin corrosion
 - Diesel exhaust fluid injector failure

- Engine running
- Prioritised Checks to Perform
 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid injector circuit for short circuit to power
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Check and install a new diesel exhaust fluid injector as required
 - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P205B-84	Reductant Tank Temperature Sensor "A" Circuit Low - Signal below allowable range	■ Purpose of the DTC ■ Monitoring of the diesel exhaust fluid level sensor (temperature) ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ Yes ■ Control Module Cavity ■ Circuit reference I_T_RAULS Signal ■ Monitor Description ■ Temperature difference below maximum limit ■ Prioritised List of Possible Causes ■ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ■ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion	■ Vehicle Conditions to enable DTC Logging Strategy ■ Cold start detection (engine off for more than 28000 seconds) ■ Cold start detection (engine off for more than 28000s, Tenv Tfuel and Tcat within 7°C) ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Contact dealer technical support ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P205B-85	Reductant Tank Temperature	■ Purpose of the DTC ■ Monitoring of	■ Vehicle Conditions to enable DTC Logging Strategy

	Sensor "A" Circuit Low - Signal above allowable range	the diesel exhaust fluid level sensor (temperature) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Temperature difference above maximum limit ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Cold start detection (engine off for more than 28000 seconds) ▪ Cold start detection (engine off for more than 28000s, Tenv Tfuel and Tcat within 7°C) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P205C-16	Reductant Tank Temperature Sensor "A" Circuit High - Circuit	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform

	voltage below threshold	level sensor (temperature) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Diesel exhaust fluid tank module temperature signal range check too low ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P205D-17	Reductant Tank Temperature Sensor "A" Circuit Intermittent/Erratic - Circuit voltage	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform

	above threshold	(temperature) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Diesel exhaust fluid tank module temperature signal range check too high ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P205E-64	Reductant Tank Temperature Sensor Circuit Intermittent/Erratic - Signal plausibility	▪ Purpose of the DTC ▪ Monitoring of the diesel exhaust fluid level sensor	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Key On ▪ DEF level greater than 15%

	failure	(temperature) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference I_T_RAULS Signal ▪ Monitor Description ▪ Diesel exhaust fluid tank module temperature plausibility, signal too low ▪ Prioritised List of Possible Causes ▪ The engine control module detected plausibility failures ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ DEF tank unfrozen ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P208A-13	Reductant Pump "A" Control Circuit/Open - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for open circuit	

- Electrical Cause
 - Yes
- Mechanical Cause
 - No
- Control Module Cavity
 - Circuit reference RAPM
- Monitor Description
 - Diesel exhaust fluid tank module - output from the ECM to DEF pump control open circuit
- Prioritised List of Possible Causes
 - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output
 - Diesel exhaust fluid tank module circuit open circuit, high resistance - output from the ECM to DEF pump control
 - Connector is disconnected, connector pin is backed out, connector pin corrosion

NOTE:

A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start

- Prioritised Checks to Perform
 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for open circuit, high resistance - output from the ECM to DEF pump control
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Contact dealer technical support
- Vehicle Conditions to enable DTC Logging Strategy
 - Engine running
 - On completion of the previous repair actions the following MUST be carried out
 - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
 - NOTE: The engine MIL is OFF (when engine ON)
 - Start engine
 - The 'inducement' warning clears automatically once

			the monitoring has executed and passed with no malfunction. This terminates the drive cycle ■ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ■ NOTE: This clears the fault storage from P2BAE / P2BAF
P208B-00	Reductant Pump "A" Control Performance/Stuck Off - No sub type information	■ Purpose of the DTC ■ To monitor diesel exhaust fluid line pressure build up for over pressure ■ Electrical Cause ■ No ■ Mechanical Cause ■ Yes ■ Control Module Cavity ■ Not applicable ■ Monitor Description ■ Diesel exhaust fluid line pressure build up failure ■ Prioritised List of Possible Causes ■ Diesel exhaust fluid line not connected ■ Diesel exhaust fluid line mechanical	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C ■ Prioritised Checks to Perform ■ Check diesel exhaust fluid line is correctly installed ■ Check diesel exhaust fluid line for mechanical integrity ■ Check diesel exhaust fluid injector for mechanical integrity

			integrity ▪ Diesel exhaust fluid injector mechanical integrity ▪ Diesel exhaust fluid injector stuck open ▪ Diesel exhaust fluid line heater frozen - restricted flow ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Check diesel exhaust fluid injector is not stuck open ▪ Check diesel exhaust fluid line heater is not frozen - restricted flow ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - monitor for the pump pressure build up ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the manufacturer approved diagnostic system carry out 'Diesel Exhaust Fluid Prime and Pressure Test' routine (0x409A) ▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ On completion of the previous repair actions the following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Drive vehicle at urban speeds approximately 50mph ▪ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the system pressure build up is	
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			initiated ▪ The monitoring is executed within 15 minutes of the driving cycle ▪ The 'inducement' warning clears automatically once the monitoring has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P208B-4B	Reductant Pump "A" Control Performance/Stuck Off - Over temperature	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for over temperature ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAPM ▪ Monitor Description ▪ Diesel exhaust fluid tank module - output from the ECM to DEF pump control over temperature ▪ Prioritised List of	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module control circuit for short circuit to power - output from the ECM to DEF pump ▪ Check for over heating of the module or evidence of over heating near to the module ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the manufacturer approved diagnostic system carry out 'Diesel Exhaust Fluid Prime and Pressure Test' routine (0x409A) ▪ Contact dealer technical support

		Possible Causes ▪ The engine control module detected an internal temperature above the expected range ▪ Diesel exhaust fluid tank module has over heated ▪ Diesel exhaust fluid tank module circuit short circuit to power - output from the ECM to DEF pump control ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P208B-64	Reductant Pump "A" Control Performance/Stuck Off - Signal plausibility failure	▪ Purpose of the DTC ▪ To monitor for the pump pressure ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid injection pump range performance ▪ Prioritised List of Possible Causes ▪ The engine control module detected	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - monitor for the pump pressure build up ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the manufacturer approved diagnostic system carry out 'Diesel Exhaust Fluid Prime and Pressure Test' routine (0x409A)

		- plausibility failures ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P208C-11	Reductant Pump "A" Control Circuit Low - Circuit short to ground	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for short circuit to ground ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAPM ▪ Monitor Description ▪ Diesel exhaust fluid tank module control short circuit to ground - output from the ECM to DEF pump ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground - output from the ECM to DEF pump ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to ground - output from the ECM to DEF pump ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	
P208D-12	Reductant Pump "A" Control Circuit High - Circuit short to battery	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - output from the ECM to DEF pump control for short circuit to power ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference RAPM ▪ Monitor Description ▪ Diesel exhaust fluid tank module control short circuit to power - output from the ECM to DEF pump	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to power - output from the ECM to DEF pump ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Prioritised List of Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to power - output from the ECM to DEF pump ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	
P208E-73	Reductant Injection Valve Stuck Closed (Bank 1 Unit 1) - Actuator stuck closed	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid injector opening / closing ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid injector is stuck closed ▪ Prioritised List of	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Check and install a new diesel exhaust fluid injector as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		Possible Causes ▪ The engine control module has not detected any motion, upon commanding the operation of a motor, solenoid or relay to open some piece of equipment ▪ Diesel exhaust fluid injector failure	
P209F-16	Reductant Tank Heater Control Circuit Performance - Circuit voltage below threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit voltage too low ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Diesel exhaust fluid heater control unit circuit open circuit, high	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid heater control unit failure	
P209F-17	Reductant Tank Heater Control Circuit Performance - Circuit voltage above threshold	- Purpose of the DTC - To monitor diesel exhaust fluid heater control unit - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid heater control unit voltage too high - Prioritised List of Possible Causes - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power - Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power,	- Vehicle Conditions to enable DTC Logging Strategy - DEF heater on - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid heater control unit as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Diesel exhaust fluid heater control unit failure	
P209F-18	Reductant Tank Heater Control Circuit Performance - Circuit current below threshold	- Purpose of the DTC - To monitor diesel exhaust fluid heater control unit - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid heater control unit current too low - Prioritised List of Possible Causes - The engine control module has measured current flow below a specified range - Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Vehicle Conditions to enable DTC Logging Strategy - DEF heater on - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new diesel exhaust fluid heater control unit as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P209F-19	Reductant Tank Heater Control Circuit Performance - Circuit current above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid heater control unit current too high ▪ Prioritised List of Possible Causes ▪ The engine control module has measured current flow above a specified range ▪ Diesel exhaust fluid heater control unit circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected,	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid heater control unit as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid heater control unit failure	
P20B9-13	Reductant Heater "A" Control Circuit/Open - Circuit open	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid line heater circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		corrosion ▪ Diesel exhaust fluid line heater failure	
P20BA-16	Reductant Heater "A" Control Circuit Performance - Circuit voltage below threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater voltage too low ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Diesel exhaust fluid line heater circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P20BA-17	Reductant Heater "A" Control Circuit Performance - Circuit voltage above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater voltage too high ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20BA-18	Reductant Heater "A" Control Circuit Performance - Circuit current	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform

	below threshold	▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater current too low ▪ Prioritised List of Possible Causes ▪ The engine control module has measured current flow below a specified range ▪ Diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20BA-19	Reductant Heater "A" Control Circuit Performance - Circuit current above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF heater on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit for short circuit to ground, short circuit to power, open

		- Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater current too high ▪ Prioritised List of Possible Causes ▪ The engine control module has measured current flow above a specified range ▪ Diesel exhaust fluid line heater circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	- circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20BB-11	Reductant Heater "A" Control Circuit Low - Circuit short to ground	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater short circuit to ground	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit for short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required

		▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid line heater circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20BC-12	Reductant Heater "A" Control Circuit High - Circuit short to battery	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid line heater ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid line heater short circuit to power ▪ Prioritised List of	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid line heater circuit for short circuit to power ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid line heater as required ▪ Using the manufacturer approved diagnostic system

		Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Diesel exhaust fluid line heater circuit short circuit to power ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid line heater failure	clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20E8-84	Reductant Pressure Too Low - Signal below allowable range	▪ Purpose of the DTC ▪ To monitor for diesel exhaust fluid line under pressure error ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ If modelled pressure falls below 2 bar system is purged and pressure is built up again. If	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C ▪ Prioritised Checks to Perform

			this happens 12 times during driving cycle error is set (Fast path due to air bubbles) ■ Prioritised List of Possible Causes ■ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ■ DEF injector dosing too high. Diesel Exhaust Fluid injection pump delivers too much DEF ■ Diesel exhaust fluid line leakage ■ Diesel exhaust fluid line air bubbles ■ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion	■ Check diesel exhaust fluid line for leakage ■ Using the manufacturer approved diagnostic system carry out diagnostic routine 0x4092 ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Contact dealer technical support ■ Vehicle Conditions to enable DTC Logging Strategy ■ DEF system pressurized ■ On completion of the previous repair actions the following MUST be carried out ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ NOTE: The engine MIL is OFF (when engine ON) ■ Drive vehicle at urban speeds approximately 50mph ■ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the system pressure build up is initiated ■ The monitoring is executed within 15 minutes of the driving cycle ■ The 'inducement' warning	
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			clears automatically once the monitoring has executed. This terminates the drive cycle - NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault storage from P2BAE / P2BAF
P20E8-85	Reductant Pressure Too Low - Signal above allowable range	- Purpose of the DTC - To monitor for diesel exhaust fluid line under pressure error - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - If modelled pressure falls below 4.5 bar the system is purged and pressure is built up again. If this occurs during a drive cycle the error is set - Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity,	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C - Prioritised Checks to Perform - Check diesel exhaust fluid line for leakage - Using the manufacturer approved diagnostic system carry out diagnostic routine 0x4092 - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short

			reported via serial data, is above a specified range ▪ DEF injector dosing too high. Diesel Exhaust Fluid injection pump delivers too much DEF ▪ Diesel exhaust fluid line leakage ▪ Diesel exhaust fluid line air bubbles ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ On completion of the previous repair actions the following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Drive vehicle at urban speeds approximately 50mph ▪ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the system pressure build up is initiated ▪ The monitoring is executed within 15 minutes of the driving cycle ▪ The 'inducement' warning clears automatically once the monitoring has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using	
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			the 'Diagnosis Menu' tab NOTE: This clears the fault storage from P2BAE / P2BAF
P20E9-84	Reductant Pressure Too High - Signal below allowable range	- Purpose of the DTC - To monitor for over pressure (slow) - Electrical Cause - No - Mechanical Cause - Yes - Control module cavity - Not applicable - Monitor Description - If modelled pressure rises above 8,5 bar system is purged and pressure is built up again. If this happens 5 times during driving cycle error is set - Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Diesel exhaust fluid line frozen / blocked - Diesel exhaust	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C - Prioritised Checks to Perform - Check diesel exhaust fluid line for blockage - Check diesel exhaust fluid injector for blockage - Check diesel exhaust fluid tank module for blockage - Check diesel exhaust fluid line is not frozen - Check diesel exhaust fluid injector is not frozen - Check diesel exhaust fluid tank module is not frozen - Vehicle Conditions to enable DTC Logging Strategy - DEF system pressurized - On completion of the previous repair actions the

			fluid line heater frozen ▪ Diesel exhaust fluid injector frozen / blocked ▪ Diesel exhaust fluid tank module frozen	following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Drive vehicle at urban speeds approximately 50mph ▪ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the system pressure build up is initiated ▪ The monitoring is executed within 15 minutes of the driving cycle ▪ The 'inducement' warning clears automatically once the monitoring has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF	
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P20E9-85	Reductant Pressure Too High - Signal above allowable range	■ Purpose of the DTC ■ To monitor for over pressure (fast) ■ Electrical Cause ■ No ■ Mechanical Cause ■ Yes ■ Control Module Cavity ■ Not applicable ■ Monitor Description ■ If modelled pressure rises above 10 bar, system is purged and pressure is built up again. If this happens 5 times during driving cycle error is set ■ Prioritised List of Possible Causes ■ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ■ Diesel exhaust fluid line frozen / blocked ■ Diesel exhaust fluid line heater frozen ■ Diesel exhaust fluid injector frozen / blocked ■ Diesel exhaust	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C ■ Prioritised Checks to Perform ■ Check diesel exhaust fluid line for blockage ■ Check diesel exhaust fluid injector for blockage ■ Check diesel exhaust fluid tank module for blockage ■ Check diesel exhaust fluid line is not frozen ■ Check diesel exhaust fluid injector is not frozen ■ Check diesel exhaust fluid tank module is not frozen ■ Vehicle Conditions to enable DTC Logging Strategy ■ DEF system pressurized ■ On completion of the previous repair actions the following MUST be carried out ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab

		fluid tank module frozen	and retest ■ NOTE: The engine MIL is OFF (when engine ON) ■ Drive vehicle at urban speeds approximately 50mph ■ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the system pressure build up is initiated ■ The monitoring is executed within 15 minutes of the driving cycle ■ The 'inducement' warning clears automatically once the monitoring has executed. This terminates the drive cycle ■ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ■ NOTE: This clears the fault storage from P2BAE / P2BAF
P20EE-00	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - No sub type information	■ Purpose of the DTC ■ Monitors SCR Nox catalyst efficiency - reduced catalyst Nox efficiency ■ The SCR device ammonia storage capability is greatly reduced or the device is missing ■ Electrical Cause	■ Vehicle Conditions to enable DTC Logging Strategy ■ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 20min (50 / 60mph) ■ Prioritised Checks to Perform ■ Damaged/Removed SCR catalyst ■ Deposits on front face of

		▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Prioritised List of Possible Causes ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20EE-04	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - System internal failures	▪ Purpose of the DTC ▪ Selective catalytic reduction - reduced catalyst Nox conversion	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the

			efficiency by passive monitoring ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Prioritised List of Possible Causes ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal	vehicle is not in DPF active regeneration and drive at urban speeds for 20min (50 / 60mph) ▪ Prioritised Checks to Perform ▪ Damaged/Removed SCR catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	
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		work SCR catalyst damage	
P20EE-92	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - Performance or incorrect operation	- Purpose of the DTC - Selective catalytic reduction - reduced catalyst Nox conversion efficiency in ammonia storage mode - The SCR device ammonia storage capability is greatly reduced or the device is missing - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions - The SCR device ammonia storage capability is greatly reduced or the device is missing - Prioritised List of	- Vehicle Conditions to enable DTC Logging strategy - Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 20min (50 / 60mph) - Prioritised Checks to Perform - Damaged/Removed SCR catalyst - Deposits on front face of SCR catalyst - Check diesel exhaust fluid injector for deposits or corrosion - Check exhaust metal work for damage - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		Possible Causes ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	
P20EE-96	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - Component internal failure	▪ Purpose of the DTC ▪ Selective catalytic reduction - reduced catalyst Nox conversion efficiency by active monitoring ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ SCR - reduced catalyst Nox conversion efficiency detected by active monitoring but not detected by active	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 20min (50 / 60mph) ▪ Prioritised Checks to Perform ▪ Damaged/Removed SCR catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		monitoring; on two consecutive occasions ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Prioritised List of Possible Causes ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	
P20EE-97	SCR NOx Catalyst Efficiency Below Threshold (Bank 1) - Component or System operation obstructed or blocked	▪ Purpose of the DTC ▪ Selective catalytic reduction - reduced catalyst Nox conversion efficiency in ammonia pressure control mode ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 30min (70 mph), with light throttle pedal transients ▪ Prioritised Checks to Perform ▪ Damaged/Removed SCR catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage

		- Cavity ▪ Not applicable ▪ Monitor Description ▪ SCR - reduced catalyst Nox conversion efficiency detected by passive monitoring but not detected by active monitoring; on two consecutive occasions ▪ The SCR device ammonia storage capability is greatly reduced or the device is missing ▪ Prioritised List of Possible Causes ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20FA-13	Reductant Pump "B" Control Circuit/Open - Circuit open	▪ Purpose of the DTC ▪ To monitor DEF back flow pump for open circuit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control module cavity	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for open circuit, high resistance - DEF back flow pump

		▪ Circuit reference RAPP ▪ Monitor Description ▪ Diesel exhaust fluid tank module - DEF back flow pump circuit open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid tank module circuit open circuit, high resistance - DEF back flow pump ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20FB-00	Reductant Pump "B" Control Performance/Stuck Off - No sub type information	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - DEF back flow pump ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance - DEF back flow pump

		- Control module cavity - Not applicable - Monitor Description - Diesel exhaust fluid tank module - DEF back flow pump system pressure can not be reduced after ignition off - Prioritised List of Possible Causes - Diesel exhaust fluid tank module - DEF back flow pump circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20FC-11	Reductant Pump "B" Control Circuit Low - Circuit short to ground	- Purpose of the DTC - To monitor DEF back flow pump for short circuit to ground - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference RAPP - Monitor Description - Diesel exhaust fluid tank module - DEF back flow pump	- Vehicle Conditions to enable DTC Logging Strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground - DEF back flow pump - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Contact dealer technical support - Using the manufacturer approved diagnostic system

		circuit short circuit to ground ■ Prioritised List of Possible Causes ■ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ■ Diesel exhaust fluid tank module circuit short circuit to ground - DEF back flow pump ■ Connector is disconnected, connector pin is backed out, connector pin corrosion	clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20FD-12	Reductant Pump "B" Control Circuit High - Circuit short to battery	■ Purpose of the DTC ■ To monitor DEF back flow pump for short circuit to power ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ No ■ Control Module Cavity ■ Circuit reference RAPP ■ Monitor Description ■ Diesel exhaust fluid tank module - DEF back flow pump	■ Vehicle Conditions to enable DTC Logging Strategy ■ Engine running ■ Prioritised Checks to Perform ■ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to power - DEF back flow pump ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Contact dealer technical support ■ Using the manufacturer approved diagnostic system

		circuit short circuit to power ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to power - DEF back flow pump ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P20FE-00	Reductant Metering Unit Performance - No sub type information	▪ Purpose of the DTC ▪ Monitors DEF reagent delivery performance to the SCR device ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ An active overdosing of the SCR is performed following a	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 20min (50 / 60mph) ▪ Prioritised Checks to Perform ▪ Damaged/Removed dosing module ▪ Dosing module stuck closed / stuck by deposits ▪ Deposits on SCR device front face ▪ Damaged/Removed SCR

		passive monitoring event (lack of apparent SCR efficiency). The reagent delivery provided by the dosing valve is monitored during this process ▪ Prioritised List of Possible Causes ▪ Defective DEF reagent delivery ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage
P20FE-97	Reductant Metering Unit Performance - Component or system operation obstructed or blocked	▪ Purpose of the DTC ▪ Selective catalytic reduction - diesel exhaust fluid delivery efficiency by active monitoring ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Selective catalytic	▪ Vehicle Conditions to enable DTC Logging strategy ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 20min (50 / 60mph) ▪ Prioritised Checks to Perform ▪ Damaged/Removed dosing module ▪ Dosing module stuck closed / stuck by deposits ▪ Deposits on SCR device front face ▪ Damaged/Removed SCR catalyst

		reduction - Nox conversion efficiency is below threshold; the diesel exhaust fluid dosing module delivery is greatly reduced ▪ Prioritised List of Possible Causes ▪ Defective DEF reagent delivery ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for deposits or corrosion ▪ Check exhaust metal work for damage
P214E-19	Reductant Pump "A" Current Too High - Circuit current above threshold	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid tank module - Supply pump ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Diesel exhaust fluid tank module - Supply pump coil resistance is not plausible	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab

		▪ Prioritised List of Possible Causes ▪ The engine control module has measured current flow above a specified range ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	and retest
P21C5-64	Reductant Level Sensor "A" Stuck - Signal plausibility failure	▪ Purpose of the DTC ▪ To monitor DEF tank level sensor signal ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ DEF tank level sensor signal is not updated ▪ Prioritised List of Possible Causes ▪ Overfilled DEF tank - Refer to D & T section 309-00C: Exhaust system and refer to functionality of level sensor (possibility that the sensor reads	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ DEF system pressurized ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check, drain and correct diesel exhaust fluid level ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Contact dealer technical

		0 if over filled) - Customer has over filled the DEF tank - The engine control module detected plausibility failures - Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	support Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P21C7-13	Reductant Control Module Power Relay/Relays Control Circuit/Open - Circuit open	- Purpose of the DTC - To monitor diesel exhaust fluid relay - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference SCRMRLY - Monitor Description - Diesel exhaust fluid relay short circuit open circuit, high resistance - Prioritised List of Possible Causes - The engine control module has determined	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance

			an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ On completion of the previous repair actions the following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault
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			storage from P2BAE / P2BAF
P21C8-11	Reductant Control Module Power Relay/Relays Control Circuit Low - Circuit short to ground	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference SCRMRLY ▪ Monitor Description ▪ Diesel exhaust fluid relay circuit short circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement for a period longer than expected or has detected a ground measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	
P21C9-12	Reductant Control Module Power Relay/Relays Control Circuit High - Circuit short to battery	▪ Purpose of the DTC ▪ To monitor diesel exhaust fluid relay ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference SCRMRLY ▪ Monitor Description ▪ Diesel exhaust fluid relay circuit short circuit to power ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a vehicle power measurement for a period longer than expected or has detected a vehicle power	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid tank module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid relay circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new diesel exhaust fluid relay as required ▪ Contact dealer technical support ▪ Using the manufacturer

		measurement when another value was expected ▪ Diesel exhaust fluid tank module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Diesel exhaust fluid relay circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Diesel exhaust fluid relay failure	approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2200-13	NOx Sensor Circuit Bank 1 Sensor 1 - Circuit open	▪ Purpose of the DTC ▪ Circuit check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for open circuit, high resistance ▪ Inspect connectors for signs

		reduction NOx sensor circuit open circuit, high resistance ▪ Prioritised List of Possible Causes ▪ The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output ▪ Pre selective catalytic reduction NOx sensor circuit open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2200-14	NOx Sensor Circuit Bank 1 Sensor 1 - Circuit short to ground or open	▪ Purpose of the DTC ▪ Circuit check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit

		▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor short circuit to ground ▪ Prioritised List of Possible Causes ▪ The engine control module has detected a ground measurement or open circuit for a period longer than expected or has detected a ground measurement or open circuit when another value was expected ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2201-28	NOx Sensor Circuit Range/Performance Bank 1 Sensor 1 - Signal bias level out of range / zero adjustment failure	▪ Purpose of the DTC ▪ Offset check of pre selective catalytic reduction NOx sensor	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a

		▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor positive / negative offset detected during overrun ▪ Prioritised List of Possible Causes ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2201-2F	NOx Sensor Circuit Range/Performance Bank 1 Sensor 1 - Signal erratic	▪ Purpose of the DTC ▪ Feedback check of pre selective catalytic reduction NOx sensor	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation

		▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor delivers unplausible signal under steady state ▪ Prioritised List of Possible Causes ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2202-16	NOx Sensor Circuit Low Bank 1 Sensor 1 - Circuit voltage below threshold	▪ Purpose of the DTC ▪ O2 range check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function

		▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor O2 signal not in range minimum ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2203-17	NOx Sensor Circuit High Bank 1 Sensor 1 - Circuit voltage	▪ Purpose of the DTC ▪ O2 range check of pre selective	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the

above threshold

catalytic
reduction NOx
sensor

- Electrical Cause
 - Yes
- Mechanical Cause
 - No
- Control Module Cavity
 - Circuit reference CAN_L_3
 - Circuit reference CAN_3_H
- Monitor Description
 - Pre selective catalytic reduction NOx sensor O2 signal not in range maximum
- Prioritised List of Possible Causes
 - The engine control module measured a voltage above a specified range but not necessarily a short circuit to power
 - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
 - Connector is disconnected, connector pin is backed out, connector pin corrosion
 - Pre selective catalytic reduction NOx

dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring)

- Prioritised Checks to Perform
 - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Check and install a new pre selective catalytic reduction NOx sensor as required
 - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		sensor failure	
P2209-01	NOx Sensor Heater Sense Circuit Range/Performance Bank 1 Sensor 1 - General electrical failure	▪ Purpose of the DTC ▪ Heater performance check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor heater unavailable ▪ Pre selective catalytic reduction NOx sensor heater is not able to reach temperature ▪ Prioritised List of Possible Causes ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor heating on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		backed out, connector pin corrosion Pre selective catalytic reduction NOx sensor failure	
P220A-01	NOx Sensor Supply Voltage Circuit Bank 1 Sensor 1 - General electrical failure	- Purpose of the DTC - Circuit check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor heater circuit error - Prioritised List of Possible Causes - Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Pre selective	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		catalytic reduction NOx sensor failure	
P220B-01	NOx Sensor Supply Voltage Circuit Bank 1 Sensor 2 - General electrical failure	▪ Purpose of the DTC ▪ Circuit check of post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor heater circuit error ▪ Prioritised List of Possible Causes ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P225C-85	NOx Sensor Performance - Signal Biased/Stuck High Bank 1 Sensor 1 - Signal above allowable range	▪ Purpose of the DTC ▪ Range check pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control module cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor signal out of range at low load positive deviation ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
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		backed out, connector pin corrosion Pre selective catalytic reduction NOx sensor failure	
P225D-84	NOx Sensor Performance - Signal Biased/Stuck Low Bank 1 Sensor 1 - Signal below allowable range	- Purpose of the DTC - Range check pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor signal out of range at low load negative deviation - Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range - Pre selective catalytic reduction NOx sensor circuit	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	
P225E-85	NOx Sensor Performance - Signal Biased/Stuck High Bank 1 Sensor 2 - Signal above allowable range	▪ Purpose of the DTC ▪ Range check post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor signal out of range at low load positive deviation ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity,	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab

		reported via serial data, is above a specified range - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	and retest
P225F-84	NOx Sensor Performance - Signal Biased/Stuck Low Bank 1 Sensor 2 - Signal below allowable range	- Purpose of the DTC - Range check post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor signal out of range at low load negative deviation	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

		▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P229E-13	NOx Sensor Circuit (Bank 1 Sensor 2) - Circuit open	▪ Purpose of the DTC ▪ Circuit check of post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the

		- Monitor Description - Post selective catalytic reduction NOx sensor circuit open circuit, high resistance - Prioritised List of Possible Causes - The engine control module has determined an open circuit via lack of bias voltage, low current flow, no change in the state of an input in response to an output - Post selective catalytic reduction NOx sensor circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	post selective catalytic reduction NOx sensor circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P229E-14	NOx Sensor Circuit (Bank 1 Sensor 2) - Circuit short to ground or open	- Purpose of the DTC - Circuit check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then

		- Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor short circuit to ground - Prioritised List of Possible Causes - The engine control module has detected a ground measurement or open circuit for a period longer than expected or has detected a ground measurement or open circuit when another value was expected - Post selective catalytic reduction NOx sensor circuit short circuit to ground, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P229F-	NOx Sensor Circuit	Purpose of the DTC	Vehicle Conditions to enable

28	Range/Performance (Bank 1 Sensor 2)Signal bias level out of range / zero adjustment failure	▪ Offset check of post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor positive / negative offset detected during overrun ▪ Prioritised List of Possible Causes ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Overrun ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P229F-2F	NOx Sensor Circuit Range/Performance (Bank 1 Sensor 2) -	▪ Purpose of the DTC ▪ Feedback check of post selective	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the

	Signal erratic	catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control module cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor delivers unplausible signal under steady state ▪ Prioritised List of Possible Causes ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P22A0-16	NOx Sensor Circuit Low (Bank 1 Sensor 2) - Circuit voltage below threshold	▪ Purpose of the DTC ▪ O2 range check of post selective catalytic reduction NOx	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after

			sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor O2 signal not in range minimum ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage below a specified range but not necessarily a short circuit to ground ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	
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P22A1-17	NOx Sensor Circuit High (Bank 1 Sensor 2) - Circuit voltage above threshold	▪ Purpose of the DTC ▪ O2 range check of post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor O2 signal not in range maximum ▪ Prioritised List of Possible Causes ▪ The engine control module measured a voltage above a specified range but not necessarily a short circuit to power ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
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		corrosion Post selective catalytic reduction NOx sensor failure	
P22A7-01	NOx Sensor Heater Sense Circuit Range/Performance (Bank 1 Sensor 2) - General electrical failure	- Purpose of the DTC - Heater performance check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor heater unavailable - Post selective catalytic reduction NOx sensor heater is not able to reach temperature - Prioritised List of Possible Causes - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor heating on - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		- Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	
P22FA-27	NOx Sensor Performance - Slow Response High too low Bank 1 Sensor 1 - Signal rate of change above threshold	- Purpose of the DTC - Dynamic check of pre selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Pre selective catalytic reduction NOx sensor slow response to dynamic transients - Prioritised List of Possible Causes - The signal transitions more quickly than is reasonably allowed - Pre selective catalytic reduction NOx sensor circuit short circuit to	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new pre selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	
P22FB-64	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1 - Signal plausibility failure	▪ Purpose of the DTC ▪ Plausibility check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor positive / negative deviation to the modelled value ▪ Prioritised List of Possible Causes ▪ The engine control module detected plausibility failures ▪ Pre selective catalytic	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	
P22FB-84	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1 - Signal below allowable range	▪ Purpose of the DTC ▪ Out of range check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor signal out of range at low load negative deviation ▪ Pre selective catalytic reduction NOx sensor signal range check is	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system

		less than 100ppm ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure	clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P22FB-85	NOx Sensor Performance - Sensing Element Bank 1 Sensor 1 - Signal above allowable range	▪ Purpose of the DTC ▪ Out of range check of pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform

			CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Pre selective catalytic reduction NOx sensor signal out of range at low load positive deviation ▪ Pre selective catalytic reduction NOx sensor signal range check is larger than 3000ppm ▪ Prioritised List of Possible Causes ▪ The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range ▪ Pre selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Pre selective catalytic reduction NOx sensor failure ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new pre selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest	
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P22FE-2A	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal stuck in range	▪ Purpose of the DTC ▪ Post selective catalytic reduction NOx sensor plausibility ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor signal is not plausible in part load ▪ Prioritised List of Possible Causes ▪ Post selective catalytic reduction NOx sensor no reaction on high NOx peak ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running. NOx peak ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
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		catalytic reduction NOx sensor failure	
P22FE-64	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal plausibility failure	▪ Purpose of the DTC ▪ Post selective catalytic reduction NOx sensor removal detection ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor signal is not plausible in part load ▪ Prioritised List of Possible Causes ▪ The engine control module detected plausibility failures ▪ Post selective catalytic reduction NOx sensor not installed into exhaust system ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit,	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Overrun ▪ Part load ▪ Prioritised Checks to Perform ▪ Check the post selective catalytic reduction NOx sensor is correctly installed into exhaust system ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	
P22FE-84	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal below allowable range	- Purpose of the DTC - Out of range check of post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor signal out of range at low load negative deviation - Post selective catalytic reduction NOx sensor signal range check is less than 100ppm - Prioritised List of Possible Causes - The engine control module	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		has determined failures where some circuit quantity, reported via serial data, is below a specified range ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	
P22FE-85	NOx Sensor Performance - Sensing Element Bank 1 Sensor 2 - Signal above allowable range	▪ Purpose of the DTC ▪ Out of range check of post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open

		catalytic reduction NOx sensor signal out of range at low load positive deviation - Post selective catalytic reduction NOx sensor signal range check is larger than 3000ppm ■ Prioritised List of Possible Causes - The engine control module has determined failures where some circuit quantity, reported via serial data, is above a specified range - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P241D-00	SCR Inducement - Forced Engine Shutdown - No sub type information	- Purpose of the DTC - Engine restart is prevented. The inducement	- Vehicle Conditions to enable DTC Logging Strategy - With the inducement countdown at 0; stop

		system countdown has run out ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ The engine restart is not allowed as the inducement system has run out of mileage ▪ Prioritised List of Possible Causes ▪ Other related DTCs ▪ The vehicle has been driven until the end of the inducement system operation allowance	engine ▪ Ignition off ▪ Power latch ▪ Ignition on (after power latch) ▪ Carry out routine 0x0406 Value 0x0F - To Reset Selective Catalyst Reductant (SCR) Inducement System Start Inhibit to allow restart from this point onwards ▪ The routine can be used each time the mileage countdown goes to 0 ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check that the DEF tank has at least 25mm of fluid (0x05C1) ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index ▪ After repair, use routine 0x0406 Value 0x0F - To Reset Selective Catalyst Reductant (SCR) Inducement System Start Inhibit to allow a temporary mileage allowance (50km / 31mi) in order to run the vehicle and the OBD system to confirm repair ▪ The inducement system will clear automatically upon acknowledgment of repair by the OBD system	
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P2470-16	Exhaust Gas Temperature Sensor Circuit Low Bank 1 Sensor 4 - Circuit voltage below threshold	▪ Purpose of the DTC ▪ Temperature sensor voltage under threshold (80mV) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference I_A_PFTS2 ▪ Circuit reference G_R_BT51 ▪ Monitor Description ▪ Exhaust temperature sensor - Post particulate filter temperature circuit continuity check. Temperature sensor voltage is checked against a minimum threshold ▪ Prioritised List of Possible Causes ▪ Exhaust gas temperature sensor - Post DPF circuit short circuit to ground ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post DPF failure	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03C8 Exhaust Gas Temperature Bank1 Sensor 3 Voltage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to ground ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post DPF as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P2471-	Exhaust Gas	▪ Purpose of the DTC	▪ Vehicle Conditions to enable

17	Temperature Sensor Circuit High Bank 1 Sensor 4 - Circuit voltage above threshold	▪ Temperature sensor voltage above threshold (4990mV) ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference I_A_PFTS2 ▪ Circuit reference G_R_BT51 ▪ Monitor Description ▪ Exhaust temperature sensor - Post particulate filter temperature circuit continuity check. Temperature sensor voltage is checked against a maximum threshold ▪ Prioritised List of Possible Causes ▪ Exhaust gas temperature sensor - Post DPF circuit short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Exhaust gas temperature sensor - Post DPF failure	DTC Logging Strategy ▪ Ignition on ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x03C8 Exhaust Gas Temperature Bank1 Sensor 3 Voltage ▪ Refer to the electrical circuit diagrams and check the exhaust gas temperature sensor - Post DPF circuit for short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new exhaust gas temperature sensor - Post DPF as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

P249C-00

Excessive Time To Enter Closed Loop Reductant Injection Control - No sub type information

- Purpose of the DTC
 - To monitor whole selective catalyst reduction system
- Electrical Cause
 - Yes
- Mechanical Cause
 - Yes
- Control Module Cavity
 - Multiple
- Monitor Description
 - Selective catalyst reduction system can not enter into dosing mode within maximum allowed time after catalyst temperature is above minimum dosing temperature
- Prioritised List of Possible Causes
 - Other related DTCs
 - Power or ground failure
 - Diesel exhaust fluid pressure too low
 - Diesel exhaust fluid pressure too high
 - Diesel exhaust fluid injection pump
 - Diesel exhaust fluid injector
 - Diesel exhaust fluid relay
 - Diesel exhaust

NOTE:

A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C

- Prioritised Checks to Perform
 - Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index
 - Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
 - Refer to the workshop manual and the battery care manual, inspect the vehicle battery and ensure it is fully charged and serviceable before performing further tests
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Refer to the electrical circuit diagrams and check sensor, relays/fuses and control module power feeds and grounds
 - Check system for correct installation of components
 - Check system mechanical

			fluid heater control unit ■ Pre selective catalytic reduction NOx sensor ■ Pre selective catalyst reduction NOx sensor module ■ Post selective catalyst reduction NOx sensor ■ Post selective catalyst reduction NOx sensor module ■ Post selective catalyst reduction soot sensor ■ Post selective catalyst reduction soot sensor module ■ Diesel sub net CAN communication bus ■ Vehicle battery failure ■ Diesel exhaust fluid level too low ■ Harness failure - Wiring integrity	and fluid integrity ■ Check system for mechanical integrity ■ Check system for leakage ■ Check system for blockage ■ Correct DEF level ■ Vehicle Conditions to enable DTC Logging Strategy ■ DEF system pressurized ■ On completion of the previous repair actions the following MUST be carried out ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ NOTE: The engine MIL is OFF (when engine ON) ■ Drive vehicle at urban speeds approximately 50mph ■ NOTE: These conditions ensure that the selective catalyst reduction catalytic converter temperature exceeds 150 °C at which the system pressure build up is initiated ■ The monitoring is executed within 36 minutes of the driving cycle ■ The 'inducement' warning clears automatically once the monitoring has executed. This terminates the drive cycle ■ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab	
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			NOTE: This clears the fault storage from P2BAE / P2BAF
P249D-00	Closed Loop Reductant Injection Control At Limit - Flow Too Low	NOTE: Selective catalytic reduction, reduced catalyst Nox conversion efficiency, dosing system cannot reach dosing setpoint. The system cannot reach target efficiency - Defective SCR catalyst (aged) - Deficient DEF reagent delivery - Diesel exhaust fluid injector partial delivery - Damaged exhaust metal work - SCR catalyst damage	- Check for damaged or removed SCR catalyst - Check for deposits on front face of SCR catalyst - Check diesel exhaust fluid injector for deposits or corrosion - Check exhaust metal work for damage - Using the manufacturer approved diagnostic system run routine - Reset Selective Catalyst Reduction Dosing Adaptation Factor - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P249E-00	Closed Loop Reductant Injection Control At Limit - Flow Too High		- Check for damaged or removed SCR catalyst - Check for deposits on front face of SCR catalyst - Check diesel exhaust fluid injector for deposits or corrosion - Check exhaust metal work for damage - Using the manufacturer approved diagnostic system run routine - Reset Selective Catalyst Reduction Dosing

		NOTE: Selective catalytic reduction, reduced catalyst Nox conversion efficiency, dosing system cannot prevent ammonia slip (ammonia is found at the back of the selective catalytic reduction catalyst) ▪ Defective SCR catalyst (aged) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector partial delivery ▪ Damaged exhaust metal work ▪ SCR catalyst damage	Adaptation Factor ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P249F-00	Excessive Time To Enter Closed Loop DPF Regeneration Control - No sub type information		▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CA Exhaust Turbine A Inlet Temperature ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage ▪ Check sensor for voltage plausibility ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 ▪ 0x03E9 Exhaust Gas

NOTE:

Diesel particulate filter temperature of the inner loop (pre turbine) did not reach target on time.
Diesel particulate filter temperature of the outer loop (post catalyst) did not reach target on time

Temperature Bank2 Sensor
2 Voltage

- Check engine control module for related DTCs and refer to relevant DTC index
- Refer to the electrical circuit diagrams and check connections are secure and wiring integrity
- Check sensors are installed correctly
- Check air induction system for leakage
- Check diesel oxidation catalyst for mechanical integrity
- Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

- Other related DTCs
- Exhaust gas temperature sensor - Pre turbine - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance
- Exhaust gas temperature sensor - Pre turbine removed from gas stream
- Exhaust gas temperature sensor - Post catalytic converter - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance
- Exhaust gas temperature sensor - Post catalytic converter removed from gas stream
- Air induction system leakage

		▪ Diesel oxidation catalyst mechanical integrity	
P24A0-00	Closed Loop DPF Regeneration Control At Limit - Temperature Too Low - No sub type information	NOTE: Diesel particulate filter temperature of the inner loop (pre turbine) is under temperature target. Diesel particulate filter temperature of the outer loop (post catalyst) is under temperature target ▪ Other related DTCs ▪ Exhaust gas temperature sensor - Pre turbine - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas temperature sensor - Pre turbine removed from gas stream ▪ Exhaust gas temperature sensor - Post catalytic converter - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas temperature sensor -	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CA Exhaust Turbine A Inlet Temperature ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage ▪ Check sensor for voltage plausibility ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Check sensors are installed correctly ▪ Check air induction system for leakage ▪ Check diesel oxidation catalyst for mechanical integrity ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		Post catalytic converter removed from gas stream ▪ Air induction system leakage ▪ Diesel oxidation catalyst mechanical integrity	
P24A1-00	Closed Loop DPF Regeneration Control At Limit - Temperature Too High - No sub type information	NOTE: Diesel particulate filter temperature of the inner loop (pre turbine) is over temperature target. Diesel particulate filter temperature of the outer loop (post catalyst) is over temperature target ▪ Other related DTCs ▪ Exhaust gas temperature sensor - Pre turbine - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance ▪ Exhaust gas temperature sensor -	▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05CA Exhaust Turbine A Inlet Temperature ▪ 0x05C9 Exhaust Turbine A Inlet Temperature Sensor Voltage ▪ Check sensor for voltage plausibility ▪ 0x03F8 Exhaust Gas Temperature Bank2 Sensor 2 ▪ 0x03E9 Exhaust Gas Temperature Bank2 Sensor 2 Voltage ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check connections are secure and wiring integrity ▪ Check sensors are installed correctly ▪ Check air induction system for leakage ▪ Check diesel oxidation catalyst for mechanical integrity

		Pre turbine removed from gas stream - Exhaust gas temperature sensor - Post catalytic converter - Wiring integrity short circuit to ground, short circuit to power, open circuit, high resistance - Exhaust gas temperature sensor - Post catalytic converter removed from gas stream - Air induction system leakage - Diesel oxidation catalyst mechanical integrity	Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
P24AE-1C	Particulate Matter Sensor Circuit - Circuit voltage out of range	- Purpose of the DTC - To monitor voltage deviations between engine control module and post selective catalyst reduction soot sensor, component protection - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Circuit reference SCRMRLY (powerfeed to	- Vehicle Conditions to enable DTC Logging Strategy - Battery voltage between 11 and 16 volts - Monitor runs only when post selective catalyst reduction soot sensor is in regeneration (requires driving for 20 minutes and visible, check 0xD940 ~785°C) - Prioritised Checks to Perform - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0xD940 Particulate Matter Sensor Meander Temperature – Raw - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for open circuit, high resistance

		post selective catalyst reduction soot sensor) - Monitor Description - Fault check to detect battery error at post selective catalyst reduction soot sensor - Prioritised List of Possible Causes - The engine control module has detected a voltage outside of the expected range, but not identified as too high or too low - CAN_L_3 circuit open circuit, high resistance - CAN_3_H circuit open circuit, high resistance - Post selective catalyst reduction soot sensor circuit open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion	- Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index
P24AE-96	Particulate Matter Sensor Circuit - Component	- Purpose of the DTC - To monitor interdigital	- Vehicle Conditions to enable DTC Logging Strategy - Monitor runs once when

internal failure

electrode post
selective catalyst
reduction soot
sensor,
component
protection

post selective catalyst
reduction soot sensor is in
regeneration (requires
driving for 20 minutes and
visible, check 0xD940
~785°C)

- Electrical Cause
 - Yes
- Mechanical Cause
 - Yes
- Control Module Cavity
 - Circuit reference CAN_L_3
 - Circuit reference CAN_3_H
 - Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor)
- Monitor Description
 - Functional IDE shunt short circuit error of post selective catalyst reduction soot sensor
- Prioritised List of Possible Causes
 - The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault
 - CAN_L_3 circuit short circuit to ground, open circuit, high

- Prioritised Checks to Perform
 - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals
 - 0xD940 Particulate Matter Sensor Meander Temperature – Raw
 - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, open circuit, high resistance
 - Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Check post selective catalyst reduction soot sensor for soot deposits on probe
 - Check post selective catalyst reduction soot sensor for water deposits on probe
 - Check and install a new post selective catalyst reduction soot sensor as required
 - Check DPF for substrate damage
 - Using the manufacturer approved diagnostic system check engine control module for related DTCs

		resistance ■ CAN_3_H circuit short circuit to ground, open circuit, high resistance ■ Post selective catalyst reduction soot sensor circuit short circuit to ground, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Post selective catalyst reduction soot sensor soot deposits on probe ■ Post selective catalyst reduction soot sensor water deposits on probe ■ Post selective catalyst reduction soot sensor failure ■ DPF damage	and refer to relevant DTC index
P24AF-29	Particulate Matter Sensor Circuit Range/Performance - Signal invalid	■ Purpose of the DTC ■ Component protection ■ Electrical Cause	■ Vehicle Conditions to enable DTC Logging Strategy ■ Ignition on for 60 seconds ■ Prioritised Checks to Perform

			▪ Yes	
			▪ Mechanical Cause ▪ No	
			▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor)	
			▪ Monitor Description ▪ Monitoring of the electrode adjustment value of the post selective catalyst reduction soot sensor	
			▪ Prioritised List of Possible Causes ▪ The value of the signal measured by the engine control module is not plausible given the operating conditions ▪ CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Post selective catalyst reduction soot	▪ Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalyst reduction soot sensor module as required ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index

		sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Post selective catalyst reduction soot sensor failure	
P24AF-96	Particulate Matter Sensor Circuit Range/Performance - Component internal failure	- Purpose of the DTC - To monitor post selective catalyst reduction soot sensor is capable of collect soot, component protection - Electrical interdigital electrode monitoring, component protection - Post selective catalyst reduction soot sensor status communication to engine control module interrupted, component protection - Electrical Cause	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on for 60 seconds - Monitor runs once when post selective catalyst reduction soot sensor is in regeneration (requires driving for 20 minutes and visible, check 0xD940 ~785°C) - Prioritised Checks to Perform - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0xD940 Particulate Matter Sensor Meander Temperature – Raw - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high

			▪ Yes	resistance	
			▪ Mechanical Cause ▪ Yes	▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance	
			▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor)	▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion	
			▪ Monitor Description ▪ Post selective catalyst reduction soot sensor removed from exhaust gas flow or blocked ▪ Functional interdigital electrode self diagnostic error of particulate ▪ Post selective catalyst reduction soot sensor signal is not valid	▪ Check post selective catalyst reduction soot sensor for blockage ▪ Check post selective catalyst reduction soot sensor for removal from exhaust gas flow	
			▪ Prioritised List of Possible Causes ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault	▪ Check and install a new post selective catalyst reduction soot sensor as required	
			▪ CAN_L_3 circuit	▪ Check DPF for substrate damage ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index	

short circuit to ground, short circuit to power, open circuit, high resistance

- CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Post selective catalyst reduction soot sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance
- Connector is disconnected, connector pin is backed out, connector pin corrosion
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
- Post selective catalyst reduction soot sensor blocked
- Post selective catalyst reduction soot sensor removal from exhaust gas flow
- Post selective catalyst reduction soot sensor failure
- DPF damage

P24B0-96

Particulate Matter Sensor Circuit Low - Component internal failure

- Purpose of the DTC
 - Electrical interdigital electrode monitoring, component protection
- Electrical Cause
 - Yes
- Mechanical Cause
 - Yes
- Control Module Cavity
 - Circuit reference CAN_L_3
 - Circuit reference CAN_3_H
 - Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor)
- Monitor Description
 - Electrical interdigital electrode negative error during voltage off of particulate sensor
- Prioritised List of Possible Causes
 - The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault
 - CAN_L_3 circuit short circuit to ground, short
- Vehicle Conditions to enable DTC Logging Strategy
 - Battery voltage between 11 and 16 volts
 - Monitor runs after ignition on
- Prioritised Checks to Perform
 - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance
 - Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance
 - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Check post selective catalyst reduction soot sensor for blockage
 - Check and install a new post selective catalyst reduction soot sensor as required
 - Check DPF for substrate damage
 - Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index

		circuit to power, open circuit, high resistance ▪ CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Post selective catalyst reduction soot sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Post selective catalyst reduction soot sensor blocked ▪ Post selective catalyst reduction soot sensor failure ▪ DPF damage	
P24B1-96	Particulate Matter Sensor Circuit High - Component internal failure	▪ Purpose of the DTC ▪ Electrical interdigital electrode monitoring, component protection ▪ Electrical Cause	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Monitor runs 500 seconds after post selective catalyst reduction soot sensor is in regeneration (requires driving for 20 minutes and visible, check 0xD940 ~785°C)

			▪ Yes	▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0xD940 Particulate Matter Sensor Meander Temperature – Raw ▪ Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check post selective catalyst reduction soot sensor for blockage ▪ Check and install a new post selective catalyst reduction soot sensor as required ▪ Check and install a new post selective catalyst reduction soot sensor module as required ▪ Check DPF for substrate damage ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index
			▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Electrical interdigital electrode negative error during voltage on of particulate sensor ▪ Prioritised List of Possible Causes ▪ The engine control module has received an indication about the component that indicates a failure e.g. an intelligent actuator or sensor is indicating an internal fault ▪ CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ CAN_3_H circuit short circuit to ground, short circuit to power, open circuit,	

		high resistance - Post selective catalyst reduction soot sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Post selective catalyst reduction soot sensor blocked - Post selective catalyst reduction soot sensor failure - Post selective catalyst reduction soot sensor module failure - DPF damage	
P24B3-00	Particulate Matter Sensor Heater Control Circuit/Open - No sub type information	- Purpose of the DTC - Electrical heater monitoring, component protection - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on for 60 seconds - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance

			▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Rationality check of heater resistance executed once after main relay wakes up ▪ Prioritised List of Possible Causes ▪ CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Post selective catalyst reduction soot sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for	▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check post selective catalyst reduction soot sensor for blockage ▪ Check and install a new post selective catalyst reduction soot sensor module as required ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index	
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		signs of water ingress, and pins for damage and/or corrosion ▪ Post selective catalyst reduction soot sensor blocked ▪ Post selective catalyst reduction soot sensor module failure	
P24B5-92	Particulate Matter Sensor Heater Control Circuit Low - Performance or incorrect operation	▪ Purpose of the DTC ▪ Electrical heater monitoring, component protection ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Electrical heater off error of particulate sensor while heater off ▪ Prioritised List of Possible Causes ▪ The engine control module has detected that the	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Battery voltage between 11 and 16 volts ▪ Monitor runs shortly after ignition on ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check post selective catalyst reduction soot sensor for blockage ▪ Check and install a new post selective catalyst reduction soot sensor module as required ▪ Using the manufacturer

			component performance is outside its expected range or operating in an incorrect way ▪ CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Post selective catalyst reduction soot sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Post selective catalyst reduction soot sensor blocked ▪ Post selective catalyst reduction soot sensor module failure	approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index	
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P24B6-92	Particulate Matter Sensor Heater Control Circuit High - Performance or incorrect operation	▪ Purpose of the DTC ▪ Electrical heater monitoring, component protection ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Electrical heater off error of particulate sensor while heater on ▪ Prioritised List of Possible Causes ▪ The engine control module has detected that the component performance is outside its expected range or operating in an incorrect way ▪ CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ CAN_3_H circuit short circuit to	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Battery voltage between 11 and 16 volts ▪ Monitor runs once when post selective catalyst reduction soot sensor is in regeneration (requires driving for 20 minutes and visible, check 0xD940 ~785°C) ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0xD940 Particulate Matter Sensor Meander Temperature – Raw ▪ Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor module circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check post selective catalyst reduction soot sensor for blockage ▪ Check and install a new post selective catalyst reduction soot sensor module as required ▪ Using the manufacturer approved diagnostic system
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		ground, short circuit to power, open circuit, high resistance - Post selective catalyst reduction soot sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Post selective catalyst reduction soot sensor blocked - Post selective catalyst reduction soot sensor module failure	check engine control module for related DTCs and refer to relevant DTC index
P24C6-00	Particulate Matter Sensor Temperature Circuit - No sub type information	- Purpose of the DTC - Electrical post selective catalyst reduction soot sensor temperature monitoring, component protection - Electrical Cause - Yes - Mechanical Cause - No	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on for 60 seconds - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the

			- Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) - Monitor Description - Post selective catalyst reduction soot sensor internal meander temperature monitoring - Prioritised List of Possible Causes - CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance - CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance - Post selective catalyst reduction soot sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin	post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalyst reduction soot sensor module as required - Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index	
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		corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion Post selective catalyst reduction soot sensor failure	
P24C6-84	Particulate Matter Sensor Temperature Circuit - Signal below allowable range	- Purpose of the DTC - Functional post selective catalyst reduction soot sensor temperature monitoring, component protection - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) - Monitor Description - Post selective catalyst reduction soot sensor monitoring of the minimum measuring range of the meander starting	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on for 60 seconds - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalyst reduction soot sensor module as required - Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index

temperature

- Prioritised List of Possible Causes
 - The engine control module has determined failures where some circuit quantity, reported via serial data, is below a specified range
 - CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance
 - CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance
 - Post selective catalyst reduction soot sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance
 - Connector is disconnected, connector pin is backed out, connector pin corrosion
Inspect connectors for signs of water ingress, and pins for damage and/or corrosion
 - Post selective catalyst reduction soot sensor failure

P24C7-62	Particulate Matter Sensor Temperature Circuit Range/Performance - Signal compare failure	▪ Purpose of the DTC ▪ Functional post selective catalyst reduction soot sensor monitoring, component protection ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Monitoring of the meander starting temperature for plausibility ▪ Prioritised List of Possible Causes ▪ Power feed to the Post selective catalyst reduction soot sensor module fell below 9 volts for longer than 0.4 seconds ▪ Post selective catalyst reduction soot sensor module circuit short circuit to	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on for 60 seconds ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalyst reduction soot sensor module as required ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index
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		ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Post selective catalyst reduction soot sensor failure	
P24C7-64	Particulate Matter Sensor Temperature Circuit Range/Performance - Signal plausibility failure	- Purpose of the DTC - Functional post selective catalyst reduction soot sensor monitoring, component protection - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) - Monitor Description - Monitoring of	- Vehicle Conditions to enable DTC Logging Strategy - Monitor runs 220seconds after post selective catalyst reduction soot sensor completes regeneration (requires driving for around 20 minutes and regeneration end visible via temperature DID D940 drops from 785°C down to ~ 200°C) - keep low rpm - Prioritised Checks to Perform - Check post selective catalyst reduction soot sensor for correct installation - Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion

			the dynamic meander temperature for plausibility ▪ Prioritised List of Possible Causes ▪ Post selective catalyst reduction soot sensor incorrectly installed ▪ Post selective catalyst reduction soot sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Post selective catalyst reduction soot sensor failure ▪ DPF outlet temperature sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ SCR catalyst blocked ▪ Exhaust system mechanical	▪ Check and install a new post selective catalyst reduction soot sensor module as required ▪ Refer to the electrical circuit diagrams and check the DPF outlet temperature sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Check SCR catalyst for blockage ▪ Check exhaust system for mechanical integrity ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index	
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		integrity	
P24D0-00	Particulate Matter Sensor Supply Voltage Circuit Low - No sub type information	▪ Purpose of the DTC ▪ To monitor voltage deviations between engine control module and post selective catalyst reduction soot sensor, component protection ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Fault check to detect battery error at post selective catalyst reduction soot sensor ▪ Prioritised List of Possible Causes ▪ CAN_L_3 circuit open circuit, high resistance ▪ CAN_3_H circuit open circuit, high resistance ▪ Post selective catalyst	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Battery voltage between 11 and 16 volts ▪ Monitor runs only when post selective catalyst reduction soot sensor is in regeneration (requires driving for 20 minutes and visible, check 0xD940 ~785°C) ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0xD940 Particulate Matter Sensor Meander Temperature – Raw ▪ Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor circuit for open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index

		reduction soot sensor circuit open circuit, high resistance Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion	
P2BA7-7B	NOx Exceedance - Empty Reagent Tank - Low fluid level	- Purpose of the DTC - To monitor diesel exhaust fluid level - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Diesel exhaust fluid tank is empty - Prioritised List of Possible Causes - The engine control module has detected that a fluid level is too low for proper operation of the system - Diesel exhaust fluid level is low	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The pressure build-up normally occurs once the selective catalyst reduction catalytic converter temperature exceeds 150 °C - Prioritised Checks to Perform - Check and correct diesel exhaust fluid level - Vehicle Conditions to enable DTC Logging Strategy - DEF system pressurized - On completion of the previous repair actions the following MUST be carried out - Refill DEF tank with at least

			5 litres DEF OR if DEF tank already full drain 2 litres of DEF ▪ Using the manufacturer approved diagnostic system run routine – To Reset Selective Catalytic Reduction Start Inhibit ▪ Drive vehicle at urban speeds approximately 50mph ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P2BA9-00	NOx Exceedance - Insufficient Reagent Quality - No sub type information	▪ Purpose of the DTC ▪ To monitor the selective catalytic reduction system for sudden reduction in catalyst Nox conversion efficiency by passive monitoring ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Selective catalytic reduction - Nox conversion efficiency is below threshold. The diesel exhaust fluid	▪ Prioritised Checks to Perform ▪ Using the manufacturer approved equipment measure DEF quality to check it conforms to specification ▪ Drain and refill DEF if not to specification ▪ Damaged/Removed SCR catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for blockage/failure ▪ Deposits on mixer ▪ Check and install a new diesel exhaust fluid injector as required ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ On completion of the previous repair actions the following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using

			quality is reduced or the selective catalytic reduction catalyst is missing - Monitor to prioritise the diesel exhaust fluid detection when a sudden drop in Nox conversion efficiency is detected - Prioritised List of Possible Causes - Diesel exhaust fluid quality is reduced - Defective SCR catalyst (missing) - Deficient DEF reagent delivery - Diesel exhaust fluid injector blockage/failure	the 'Diagnosis Menu' tab and retest - NOTE: Upon DTC(s) fault clear, the warning message changes to "No Restart Possible in XXX miles. Diesel Exhaust Fluid Dosing Malfunction" - Using the manufacturer approved diagnostic system run routine – To Reset Selective Catalyst Reductant (SCR) Quality Monitor - NOTE: The routine resets the DEF quality values AND preconditions the monitoring for the driving procedure. If after using the routine, the message "No Restart Possible in XXX miles. Diesel Exhaust Fluid Dosing Malfunction" has cleared it is advised to operate a driving cycle as follows to ensure DEF to specification is correctly injected into the system; but not necessary. If the warning message remains follow the driving cycle - Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors are active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds for 20minutes (50 / 60mph) and light load - NOTE: These conditions ensure a rapid execution of the monitoring by keeping exhaust temperature under 300 °C (Eu_A/Eu_B1) The monitoring is executed within 15 minutes after the NOx sensor(s) become active - The 'inducement' warning	
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			clears automatically once the monitoring has executed. This terminates the drive cycle - NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault storage from P2BAE / P2BAF
P2BA9-67	NOx Exceedance - Insufficient Reagent Quality - Signal incorrect after event	- Purpose of the DTC - To monitor the selective catalytic reduction system for minimum catalyst Nox conversion efficiency by passive monitoring not met after function reset - Electrical Cause - No - Mechanical Cause - Yes - Control Module Cavity - Not applicable - Monitor Description - Selective catalytic reduction - the minimum conversion efficiency of the catalytic device is not met after function reset - The diesel	- Prioritised Checks to Perform - Using the manufacturer approved equipment measure DEF quality to check it conforms to specification - Drain and refill DEF if not to specification - Damaged/Removed SCR catalyst - Deposits on front face of SCR catalyst - Check diesel exhaust fluid injector for blockage/failure - Deposits on mixer - Check and install a new diesel exhaust fluid injector as required - Vehicle Conditions to enable DTC Logging Strategy - On completion of the previous repair actions the following MUST be carried out - NOTE: Upon DTC(s) fault clear, the warning message changes to "No Restart Possible in XXX miles. Diesel Exhaust Fluid Dosing Malfunction"

			exhaust fluid quality is not meeting manufacturer specifications or the SCR catalyst is removed ■ Prioritised List of Possible Causes ■ Diesel exhaust fluid quality is reduced ■ Defective SCR catalyst (missing) ■ Deficient DEF reagent delivery ■ Diesel exhaust fluid injector blockage/failure	■ Using the manufacturer approved diagnostic system run routine – To Reset Selective Catalyst Reductant (SCR) Quality Monitor ■ Using the manufacturer approved diagnostic system run routine - Reset Selective Catalyst Reduction Dosing Adaptation Factor, in order to cancel any dosing adaptation that may have occurred. Inspect that no warning remains on the instrument cluster ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ NOTE: The routine resets the DEF quality values AND preconditions the monitoring for the driving procedure ■ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox sensors are active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds (50 / 60mph) and light load until warnings clear ■ NOTE: These conditions ensure a rapid execution of the monitoring by keeping exhaust temperature under 300 °C ■ (Eu_A/Eu_B1/Eu_B2/NAS_B) This monitoring is lengthy to execute; 6 conversion efficiencies must be calculated by the engine control module to test the monitoring ■ (Eu_A/Eu_B1/Eu_B2/NAS_B) If the current drive cycle exceeds 25 minutes, stop engine and restart drive	
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			cycle for 25 minutes which should heal the fault ▪ The 'inducement' warning clears automatically once the monitoring has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P2BA9-92	NOx Exceedance - Insufficient Reagent Quality - Performance or incorrect operation	▪ Purpose of the DTC ▪ To monitor the selective catalytic reduction system for minimum catalyst Nox conversion efficiency by passive monitoring not met after function reset ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Selective catalytic reduction - the minimum conversion efficiency of the catalytic device	▪ Prioritised Checks to Perform ▪ Using the manufacturer approved equipment measure DEF quality to check it conforms to specification ▪ Drain and refill DEF if not to specification ▪ Damaged/Removed SCR catalyst ▪ Deposits on front face of SCR catalyst ▪ Check diesel exhaust fluid injector for blockage/failure ▪ Deposits on mixer ▪ Check and install a new diesel exhaust fluid injector as required ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ On completion of the previous repair actions the following MUST be carried out ▪ NOTE: Upon DTC(s) fault clear, the warning message changes to "No Restart"

			is not met after function reset ▪ The diesel exhaust fluid quality is not meeting manufacturer specifications or the SCR catalyst is removed ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid quality is reduced ▪ Defective SCR catalyst (missing) ▪ Deficient DEF reagent delivery ▪ Diesel exhaust fluid injector blockage/failure	Possible in XXX miles. Diesel Exhaust Fluid Dosing Malfunction" ▪ Using the manufacturer approved diagnostic system run routine – To Reset Selective Catalyst Reductant (SCR) Quality Monitor ▪ Using the manufacturer approved diagnostic system run routine - Reset Selective Catalyst Reduction Dosing Adaptation Factor, in order to cancel any dosing adaptation that may have occurred. Inspect that no warning remains on the instrument cluster ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The routine resets the DEF quality values AND preconditions the monitoring for the driving procedure. If after using the routine, the message "No Restart Possible in XXX miles. Diesel Exhaust Fluid Dosing Malfunction" has cleared it is advised to operate a driving cycle as follows to ensure DEF to specification is now correctly injected into the system ; but not necessary. If the warning message remains follow the driving cycle ▪ If after this procedure, warnings are still visible on the instrument cluster; Using the manufacturer approved diagnostic system run application - SCR Monitoring Application ▪ Drive vehicle at urban speeds until SCR catalyst warm and exhaust Nox
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			sensors are active. Ensure the vehicle is not in DPF active regeneration and drive at urban speeds (50 / 60mph) and light load until warnings clear ▪ NOTE: These conditions ensure a rapid execution of the monitoring by keeping exhaust temperature under 300 °C ▪ (NAS_A/Eu_B2/NAS_B) The monitoring is executed within 15 minutes after the NOx sensor(s) become active ▪ (Eu_A/Eu_B1) This monitoring is lengthy to execute; 6 conversion efficiencies must be calculated by the engine control module to test the monitoring ▪ (Eu_A/Eu_B1) If the current drive cycle exceeds 25 minutes, stop engine and restart drive cycle for 25 minutes which should heal the fault ▪ The 'inducement' warning clears automatically once the monitoring has executed. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle: the malfunction has re-occurred ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE / P2BAF
P2BAE-00	NOx Exceedance - NOx Control	▪ Purpose of the DTC ▪ Stage 1 This	▪ Vehicle Conditions to enable DTC Logging Strategy

	Monitoring System - No sub type information	monitor shows at the first level of the driver advisory message - DEF low - Electrical Cause - No - Mechanical Cause - No - Control Module Cavity - Not applicable - Monitor Description - Low diesel exhaust fluid level - Prioritised List of Possible Causes - Overfilled DEF tank - Refer to D & T section 309- 00C: Exhaust system and refer to functionality of level sensor (possibility that the sensor reads 0 if over filled) - Customer has over filled the DEF tank - Diesel exhaust fluid level is low - Other related DTCs	- Ignition on - The fault path responsible for the inducement level may require driving cycle conditions - Prioritised Checks to Perform - Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals 0x05C1 Reductant Tank Fluid Level - Check, drain and correct diesel exhaust fluid level - Check engine control module for related DTCs and refer to relevant DTC index - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest - The inducement system will clear automatically upon acknowledgment of repair by the OBD system
P2BAE-02	NOx Exceedence - NOx control monitoring system - General signal failure	- Purpose of the DTC - Stage 2 This monitor shows at the second level of the driver advisory message - DEF low - Refill DEF now - Electrical Cause	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on - The fault path responsible for the inducement level may require driving cycle conditions - Prioritised Checks to Perform - Diagnosis of this DTC may

		▪ No ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ Low diesel exhaust fluid level ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid level is low ▪ Other related DTCs	require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check and correct diesel exhaust fluid level ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ The inducement system will clear automatically upon acknowledgment of repair by the OBD system
P2BAF-00	NOx System Driver Inducement Active - No sub type information	▪ Purpose of the DTC ▪ The DTC is set at first, second and third levels of final driver warning for inducement system ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ To monitor for low diesel exhaust fluid level ▪ To monitor diesel exhaust fluid quality ▪ To monitor selective catalyst	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on ▪ The fault path responsible for the inducement level may require driving cycle conditions ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check and correct diesel exhaust fluid level ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab

		reduction system hardware ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid level is low ▪ Other related DTCs	and retest ▪ The inducement system will clear automatically upon acknowledgment of repair by the OBD system
P2BAF-02	NOx System Driver Inducement Active - General signal failure	▪ Purpose of the DTC ▪ The DTC is set at second, level of final driver warning for inducement system ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ To monitor for low diesel exhaust fluid level ▪ To monitor diesel exhaust fluid quality ▪ To monitor selective catalyst reduction system hardware ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid level is low ▪ Other related DTCs	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on ▪ The fault path responsible for the inducement level may require driving cycle conditions ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check and correct diesel exhaust fluid level ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ The inducement system will clear automatically upon acknowledgment of repair by the OBD system
P2BAF-04	NOx System Driver Inducement Active - System internal failures	▪ Purpose of the DTC ▪ The DTC is set at third, level of final driver	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on

		warning for inducement system ▪ Electrical Cause ▪ No ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Not applicable ▪ Monitor Description ▪ To monitor for low diesel exhaust fluid level ▪ To monitor diesel exhaust fluid quality ▪ To monitor selective catalyst reduction system hardware ▪ Prioritised List of Possible Causes ▪ Diesel exhaust fluid level is low ▪ Other related DTCs	▪ The fault path responsible for the inducement level may require driving cycle conditions ▪ Prioritised Checks to Perform ▪ Diagnosis of this DTC may require using the manufacturer approved diagnostic system check datalogger signals ▪ 0x05C1 Reductant Tank Fluid Level ▪ Check and correct diesel exhaust fluid level ▪ Check engine control module for related DTCs and refer to relevant DTC index ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ The inducement system will clear automatically upon acknowledgment of repair by the OBD system
U029D-87	Lost Communication With NOx Sensor "A" - Missing message	▪ Purpose of the DTC ▪ To check CAN communication between engine control module and pre selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start

			▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Communication error between engine control module and pre selective catalytic reduction NOx sensor ▪ Prioritised List of Possible Causes ▪ The engine control module has not received the expected CAN signal from the pre selective catalytic reduction NOx sensor / diesel sub net CAN within the specified time interval ▪ Pre and post NOx sensor connectors are crossed ▪ Pre selective catalytic reduction NOx sensor / diesel sub net CAN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	▪ Prioritised Checks to Perform ▪ Check pre and post NOx sensor connectors are not crossed ▪ Refer to the electrical circuit diagrams and check the pre selective catalytic reduction NOx sensor / diesel sub net CAN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ On completion of the previous repair actions the following MUST be carried out ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ▪ NOTE: The engine MIL is OFF (when engine ON) ▪ Start engine ▪ The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle ▪ NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab ▪ NOTE: This clears the fault storage from P2BAE /	
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			P2BAF
U029E-87	Lost Communication With NOx Sensor "B" - Missing message	■ Purpose of the DTC ■ To check CAN communication between engine control module and post selective catalytic reduction NOx sensor ■ Electrical Cause ■ Yes ■ Mechanical Cause ■ Yes ■ Control Module Cavity ■ Circuit reference CAN_L_3 ■ Circuit reference CAN_3_H ■ Monitor Description ■ Communication error between engine control module and post selective catalytic reduction NOx sensor ■ Prioritised List of Possible Causes ■ The engine control module has not received the expected CAN signal from the post selective catalytic reduction NOx sensor / diesel sub net CAN within the specified time interval ■ Pre and post NOx sensor	NOTE: A fault clear is NOT sufficient to clear the vehicle 'inducement' system warnings. The 'inducement' warnings automatically clear ONLY upon confirmation of the repair by the vehicle OBD system. The monitoring MUST be executed. The electrical checks normally occur once ignition is on or after engine start ■ Prioritised Checks to Perform ■ Check pre and post NOx sensor connectors are not crossed ■ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor / diesel sub net CAN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Vehicle Conditions to enable DTC Logging Strategy ■ Engine running ■ On completion of the previous repair actions the following MUST be carried out ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest ■ NOTE: The engine MIL is

		connectors are crossed - Post selective catalytic reduction NOx sensor / diesel sub net CAN circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion	OFF (when engine ON) - Start engine - The 'inducement' warning clears automatically once the monitoring has executed and passed with no malfunction. This terminates the drive cycle - NOTE: If the engine MIL shows during the driving cycle, then the malfunction has re-occurred. Further repair is required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab - NOTE: This clears the fault storage from P2BAE / P2BAF
U02A3-13	Lost Communication With PM Sensor - Circuit open	- Purpose of the DTC - Diagnostic communication to engine control module interrupted, component protection - Electrical Cause - Yes - Mechanical Cause - No - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) - Monitor Description	- Vehicle Conditions to enable DTC Logging Strategy - Ignition on for 60 seconds - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, open circuit, high resistance - Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor module circuit for short circuit to ground, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check post selective catalyst reduction soot sensor for blockage - Check and install a new post selective catalyst reduction

		■ Post selective catalyst reduction soot sensor module not connected ■ Prioritised List of Possible Causes ■ CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Post selective catalyst reduction soot sensor module circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Post selective catalyst reduction soot sensor blocked ■ Post selective catalyst reduction soot sensor failure	soot sensor as required ■ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index	
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U02A3-87	Lost Communication With PM Sensor - Missing message	▪ Purpose of the DTC ▪ Interrupted CAN communication, component protection ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Post selective catalyst reduction soot sensor module lost communication ▪ Prioritised List of Possible Causes ▪ CAN_L_3 circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ CAN_3_H circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Post selective catalyst reduction soot sensor module circuit short	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on for 60 seconds ▪ Prioritised Checks to Perform ▪ Refer to the electrical circuit diagrams and check the CAN_L_3 and CAN_3_H circuit for short circuit to ground, open circuit, high resistance ▪ Refer to the electrical circuit diagrams and check the post selective catalyst reduction soot sensor module circuit for short circuit to ground, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check post selective catalyst reduction soot sensor for blockage ▪ Check and install a new post selective catalyst reduction soot sensor as required ▪ Using the manufacturer approved diagnostic system check engine control module for related DTCs and refer to relevant DTC index
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		circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Post selective catalyst reduction soot sensor blocked - Post selective catalyst reduction soot sensor failure	
U02A5-87	Lost Communication with Reductant Heater Control Module - Missing message	- Purpose of the DTC - To check CAN communication between engine control module and diesel exhaust fluid heater control unit - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module Cavity - Circuit reference CAN_L_3 - Circuit reference CAN_3_H - Monitor Description - Communication error between	- Vehicle Conditions to enable DTC Logging Strategy - Engine running - Prioritised Checks to Perform - Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit / diesel sub net CAN circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		engine control module and diesel exhaust fluid heater control unit ▪ Prioritised List of Possible Causes ▪ The engine control module has not received the expected CAN signal from the diesel exhaust fluid heater control unit / diesel sub net CAN within the specified time interval ▪ Diesel exhaust fluid heater control unit / diesel sub net CAN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	
U04A4-02	Invalid Data Received From PM Sensor - General signal failure	▪ Purpose of the DTC ▪ Incorrect CAN communication, component protection ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ No ▪ Control Module Cavity ▪ Circuit reference CAN_L_3	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Ignition on ▪ Prioritised Checks to Perform ▪ Check and install a new post selective catalyst reduction soot sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		▪ Circuit reference CAN_3_H ▪ Circuit reference SCRMRLY (powerfeed to post selective catalyst reduction soot sensor) ▪ Monitor Description ▪ Invalid data received from post selective catalyst reduction soot sensor ▪ Prioritised List of Possible Causes ▪ Post selective catalyst reduction soot sensor failure	
U05A6-08	Invalid Data Received from Reductant Heater Control Module - Bus signal / message failures	▪ Purpose of the DTC ▪ To check CAN communication between engine control module and diesel exhaust fluid heater control unit ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control module cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Communication error between engine control module and diesel exhaust	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ Engine running ▪ Prioritised Checks to Perform ▪ Check diesel exhaust fluid heater control unit / diesel sub net CAN for related DTCs and refer to relevant DTC index ▪ Refer to the electrical circuit diagrams and check the diesel exhaust fluid heater control unit / diesel sub net CAN circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest

		fluid heater control unit ▪ Prioritised List of Possible Causes ▪ The engine control module has not received the expected CAN signal from the diesel exhaust fluid heater control unit / diesel sub net CAN within the specified time interval ▪ Diesel exhaust fluid heater control unit / diesel sub net CAN circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion	
U3009-00	Control Module Ground "B" - No sub type information	▪ Purpose of the DTC ▪ Removal detection post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes ▪ Mechanical Cause ▪ Yes ▪ Control Module Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ▪ Prioritised Checks to Perform ▪ Check post selective catalytic reduction NOx sensor is correctly installed

		CAN_3_H - Monitor Description - Post selective catalytic reduction NOx sensor signal in part load not plausible - Prioritised List of Possible Causes - Post selective catalytic reduction NOx sensor not installed in the exhaust system - Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance - Connector is disconnected, connector pin is backed out, connector pin corrosion - Post selective catalytic reduction NOx sensor failure	in the exhaust system - Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance - Inspect connectors for signs of water ingress, and pins for damage and/or corrosion - Check and install a new post selective catalytic reduction NOx sensor as required - Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U300C-00	Ignition Input Off/On/Start - No sub type information	- Purpose of the DTC - Removal detection post selective catalytic reduction NOx sensor - Electrical Cause - Yes - Mechanical Cause - Yes - Control Module	- Vehicle Conditions to enable DTC Logging Strategy - NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring)

		- Cavity ▪ Circuit reference CAN_L_3 ▪ Circuit reference CAN_3_H ▪ Monitor Description ▪ Post selective catalytic reduction NOx sensor signal in part load not plausible ▪ Prioritised List of Possible Causes ▪ Post selective catalytic reduction NOx sensor not installed in the exhaust system ▪ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ▪ Connector is disconnected, connector pin is backed out, connector pin corrosion ▪ Post selective catalytic reduction NOx sensor failure	▪ Prioritised Checks to Perform ▪ Check post selective catalytic reduction NOx sensor is correctly installed in the exhaust system ▪ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ▪ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ▪ Check and install a new post selective catalytic reduction NOx sensor as required ▪ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
U300F-00	Ignition Input Accessory - No sub type information	▪ Purpose of the DTC ▪ Removal detection post selective catalytic reduction NOx sensor ▪ Electrical Cause ▪ Yes	▪ Vehicle Conditions to enable DTC Logging Strategy ▪ NOx sensor released, the dew point detection function is initiated (after engine start there is a strategy to avoid activation if water is present in the exhaust, the function

		■ Mechanical Cause ■ Yes ■ Control Module Cavity ■ Circuit reference CAN_L_3 ■ Circuit reference CAN_3_H ■ Monitor Description ■ Post selective catalytic reduction NOx sensor signal in part load not plausible ■ Prioritised List of Possible Causes ■ Post selective catalytic reduction NOx sensor not installed in the exhaust system ■ Post selective catalytic reduction NOx sensor circuit short circuit to ground, short circuit to power, open circuit, high resistance ■ Connector is disconnected, connector pin is backed out, connector pin corrosion ■ Post selective catalytic reduction NOx sensor failure	integrates the warmth quantity seen by the sensor and then releases it, then the sensor is heated and starts measuring) ■ Prioritised Checks to Perform ■ Check post selective catalytic reduction NOx sensor is correctly installed in the exhaust system ■ Refer to the electrical circuit diagrams and check the post selective catalytic reduction NOx sensor circuit for short circuit to ground, short circuit to power, open circuit, high resistance ■ Inspect connectors for signs of water ingress, and pins for damage and/or corrosion ■ Check and install a new post selective catalytic reduction NOx sensor as required ■ Using the manufacturer approved diagnostic system clear all stored DTCs using the 'Diagnosis Menu' tab and retest
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For a complete list of all diagnostic trouble codes that could be logged on this vehicle, please refer to Section 100-00. Engine Control Module